HANOI UNIVERSITY
FACULTY OF MANAGEMENT AND TOURISM

MACROECONOMICS
TUTORIAL EXERCISES

61ECO2MAC

Spring 2026

TUT 1

CASE STUDY

ADB lowers Thailand's GDP growth forecast

In its latest Asian Development Outlook (ADO) report, ADB projects Thai GDP to expand 2% this year, down from 2.9% forecast in April, with the economy slowing in the second half as front-loaded shipments to the US end.

The Thai economy is projected to slow in the second half of 2025 and next year as US tariff hikes reduce exports, weaken private consumption, and reduce private investment, according to ADB. (Photo: Bangkok Post)

Bangkok (VNA) - The Asian Development Bank (ADB) has lowered its forecast for Thailand's GDP growth for 2025 and the following year, given export deceleration, a slow recovery of foreign tourist arrivals, as well as weakened private consumption and investments.

In its latest Asian Development Outlook (ADO) report, the bank projects Thai GDP to expand 2% this year, down from 2.9% forecast in April, with the economy slowing in the second half as front-

loaded shipments to the US end. Economic growth is expected to slow further to 1.6% in 2026, down from its previous prediction of 2.9%. According to the report, the 19% US tariff on exports will slow shipments, particularly of electrical equipment, machinery, metals, processed food and vehicles. Small businesses may find it difficult to adjust to the tariffs compared with their US counterparts, which often benefit from larger operations and lower production costs. Meanwhile, a sluggish recovery of tourist arrivals together with a weakened global economy is dampening the tourism sector. The forecast for foreign tourist arrivals was downgraded from 39.5 million in April to 34 million, primarily due to fewer tourists from China. Increased competition from other regional destinations is also contributing to the projected slowdown. Many countries in the region are spending heavily to attract international visitors by developing tourism infrastructure and offering competitive costs. Weakened currencies in some countries also make them more affordable to foreign tourists, ADB noted./.

Source: https://en.vietnamplus.vn/adb-lowers-thailands-gdp-growth-forecast-post329675.vnp

1. Using the standard GDP formula, identify which specific variables are being impacted according to the article and explain how a "19% US tariff on exports" would mathematically affect Thailand's total GDP. Show your work in the article
2. When international organizations like the ADB or World Bank forecast "GDP growth," are they typically referring to **Nominal GDP** or **Real GDP**? Why is it essential for economists to use "Real" values when comparing Thailand's 2025 performance to its 2024 performance?
3. In the context of GDP accounting, is the money spent by a foreign tourist in Thailand classified as a **Consumption (C)** expenditure or an **Export (X)**? How does a "sluggish recovery of tourist arrivals" specifically impact the "Net Exports" component of Thailand's national accounts?

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. Which of the following is not a question that macroeconomists address?

a.	Why is average income high in some countries while it is low in others?
b.	Why does the price of oil rise when war erupts in the Middle East?
c.	Why do production and employment expand in some years and contract in others?
d.	Why do prices rise rapidly in some periods of time while they are more stable in other periods?

____ 2. Because every transaction has a buyer and a seller,

a.	GDP is more closely associated with an economy's income than it is with an economy's expenditure.
b.	every transaction contributes equally to an economy's income and to its expenditure.
c.	the number of firms must be equal to the number of households in a simple circular-flow diagram.
d.	firms' profits are necessarily zero in a simple circular-flow diagram.

____ 3. Which of the following is included in GDP?

a.	medical marijuana purchased from a government-run pharmacy by a glaucoma patient
b.	recreational marijuana purchased from a drug dealer by a college student
c.	recreational marijuana produced and consumed by a man in his attic
d.	All of the above are included in GDP.

____ 4. Spots, Inc. produces ink and sells it to Write on Target, which makes pens. The ink produced by Spots, Inc. is called

a.	an inventory good.
b.	a transitory good.
c.	a preliminary good.
d.	an intermediate good.

____ 5. Suppose there are only two firms in an economy: Rolling Rawhide produces rawhide and sells it to Chewy Chomp, Inc., which uses the rawhide to produce and sell dog chews. With each $2 of rawhide that it buys from Rolling Rawhide, Chewy Chomp, Inc. produces a dog chew and sells it for $5. Neither firm had any inventory at the beginning of 2007. During that year, Rolling Rawhide produced enough rawhide for 1000 dog chews. Chewy Chomp, Inc. bought 75% of that rawhide for $1500 and promised to buy the remaining 25% for $500 in 2008. Chewy Chomp, Inc. produced 750 dog chews during 2007 and sold each one during that year for $5. What was the economy's GDP for 2007?

a.	$3750
b.	$4250
c.	$5250
d.	$5750

____ 6. The total sales of all firms in the economy for a year

a.	equals GDP for the year.
b.	is larger than GDP for the year.
c.	is smaller than GDP for the year.
d.	Any of the above is possible.

____ 7. Which of the following domestically produced items is not included in GDP?

a.	a bottle of shampoo
b.	a hairdryer
c.	a haircut
d.	All of the above are included in GDP.

____ 8. Which of the following is included in U.S. GDP?

a.	the value of production by an American working in Morocco
b.	the value of production by a Moroccan working in the U.S.
c.	Both (a) and (b) are correct.
d.	Neither (a) nor (b) is correct.

____ 9. Gina, a U.S. citizen, works only in Germany. The value of Gina's production is included in

a.	U.S. GDP and German GDP.
b.	U.S. GDP and German GNP.
c.	U.S. GNP and German GDP.
d.	U.S. GNP and German GNP.

____ 10. In the national income accounts, depreciation is called

a.	"consumption of fixed capital."
b.	"negative investment."
c.	"capital taxation."
d.	"loss due to wear."

____ 11. Which of the following is not an example of a durable good?

a.	a refrigerator
b.	an automobile
c.	a business suit
d.	a furnace

____ 12. What word do economists use to refer to the purchase of goods that will be used in the future to produce more goods and services?

a.	capital
b.	consumption
c.	investment
d.	costs

____ 13. Which of the following items is the one type of household expenditure that is categorized as investment rather than consumption?

a.	spending on education
b.	the purchase of stocks and bonds
c.	the purchase of a new house
d.	the purchase of durable goods such as stoves and washing machines

____ 14. If net exports is a negative number for a particular year, then

a.	the value of firms' inventories declined over the course of the year.
b.	consumption exceeded the sum of investment and government purchases during the year.
c.	the value of goods sold to foreigners exceeded the value of foreign goods purchased during the year.
d.	the value of foreign goods purchased exceeded the value of goods sold to foreigners during the year.

____ 15. In the economy of Wrexington in 2008, exports were $200, GDP was $2000, government purchases were $300, imports were $130, and investment was $400. What was Wrexington's consumption in 2008?

a.	$970
b.	$1230
c.	$1370
d.	$1630

____ 16. Changes in real GDP reflect

a.	only changes in prices.
b.	only changes in the amounts being produced.
c.	both changes in prices and changes in the amounts being produced.
d.	neither changes in prices nor changes in the amounts being produced.

____ 17. If real GDP doubles and the GDP deflator doubles, then nominal GDP

a.	remains constant.
b.	doubles.
c.	triples.
d.	quadruples.

____ 18. Suppose an economy's production consists only of corn and soybeans. In 2005, 20 bushels of corn are sold at $4 per bushel and 10 bushels of soybeans are sold at $2 per bushel. In 2004, the price of corn was $2 per bushel and the price of soybeans was $1 per bushel. Using 2004 as the base year, it follows that, for 2005,

a.	nominal GDP is $50, real GDP is $100, and the GDP deflator is 50.
b.	nominal GDP is $50, real GDP is $100, and the GDP deflator is 200.
c.	nominal GDP is $100, real GDP is $50, and the GDP deflator is 50.
d.	nominal GDP is $100, real GDP is $50, and the GDP deflator is 200.

Table 23-5

The table below contains data for the country of Batterland, which produces only waffles and pancakes. The base year is 2006.

Year	Price of Waffles	Quantity of Waffles	Price of Pancakes	Quantity of Pancakes
2005	$2	100	$1	100
2006	$2	120	$2	150
2007	$3	150	$3	200
2008	$4	180	$3	220

____ 19. Refer to Table 23-5. In 2008, Batterland's real GDP was

a.	$540.
b.	$800.
c.	$930.
d.	$1380.

____ 20. Suppose that twenty-five years ago a country had nominal GDP of $1,000, a GDP deflator of 200, and a population of 100. Today it has nominal GDP of $3,000, a GDP deflator of 400, and population of 150. What happened to the real GDP per person?

a.	It more than doubled.
b.	It increased, but it less than doubled.
c.	It was unchanged.
d.	It decreased.

TUT 2

CASE STUDY

Indonesia's productivity gap threatens long-term economic growth

- **December 15, 2025 21:33 GMT+700**

Manpower Minister Yassierli opens the Indonesian Forum on Labour Productivity (IFLP) themed "Gig Workers: Flexibility and Vulnerability from Multiple Perspectives" in Jakarta on Tuesday (25/11/2025). (ANTARA/HO–Ministry of Manpower of the Republic of Indonesia)

Jakarta (ANTARA) - While the Indonesian government focuses on physical investment, downstreaming, and infrastructure development, labor force productivity remains a crucial factor in determining how effectively capital, technology, and labor generate added value for the national economy.

Productivity is often dubbed a "silent investment" because its impact is not always as visible as the construction of roads, ports, or industrial estates. Yet, it serves as the foundation for a country's long-term growth and competitiveness.

Without productivity gains, increased investment risks merely expanding the scale of the economy without being followed by a rise in value added per worker, making it difficult for national industrial competitiveness to improve sustainably.

Statistics Indonesia (BPS) data show that Indonesia's labor force as of February 2025 reached 153.05 million people, with 145.77 million employed, while the open unemployment rate stood at 4.76 percent.

These figures indicate that Indonesia's job market is relatively stable in quantitative terms. However, they do not fully reflect job quality or the overall level of labor productivity.

Manpower Minister Yassierli noted that the proportion of formal workers in Indonesia remains at around 39 percent, indicating that the majority of the workforce is employed in the informal sector.

The informal sector generally faces limitations in technology adoption, business management, and workforce skill development, resulting in productivity levels that tend to be lower than those in the formal sector. Overall, these conditions point to quality challenges in Indonesia's labor structure, as reflected in regional productivity comparisons. Based on data from the Ministry of Manpower, Indonesia's labor productivity in 2022 stood at around US$26.6 thousand per worker. This figure remains below the ASEAN average of approximately US$30.2 thousand per worker, indicating Indonesia's challenge in increasing value added per worker at the regional level.

Labor productivity per worker is a key indicator in economic analysis, as it describes the economic value generated by each worker within a specific period. From a macroeconomic perspective, growth supported by productivity improvements is generally considered more sustainable than growth that relies solely on an increase in the number of workers. Speaking at the Indonesia Productivity Summit 2025, Yassierli asserted that productivity is the key to strengthening Indonesia's competitiveness amid intensifying global competition. He emphasized that Indonesia's productivity gap is evident not only at the ASEAN level but also when compared with countries that have undertaken industrial transformation and human resource development earlier.

Labor productivity among members of the Organization for Economic Co-operation and Development (OECD) ranges from tens of thousands of US dollars to nearly US$100,000 per worker per year. Meanwhile, productivity levels in upper-middle-income countries stand at around US$40,000 to US$50,000 per worker annually. The minister stressed that productivity plays a crucial role in industries' ability to adapt to technological transformation, global trade dynamics, and rising demands for efficiency.

Amid the transition toward a digital and green economy, workforce demand is shifting rapidly and becoming more complex. Without consistent efforts to enhance productivity, economic growth risks being insufficient to sustainably drive an increase in per capita income. This places productivity not merely as a labor issue, but as an integral component of the national strategy to build competitiveness.

Challenges and policy direction

Yassierli assessed that the challenges in Indonesia's productivity are structural in nature, especially related to the educational attainment of the country's labor force.
Approximately 85–86 percent of the Indonesian workforce has an educational attainment of high school (SMA) or vocational high school (SMK),
Meanwhile, industries continue to demand more complex skills driven by digitalization and automation.
This situation coincides with the high proportion of informal workers, which reaches around 60 percent.
This structure limits the labor force productivity from improving rapidly because access to training, technology, and upskilling is not yet evenly distributed.
In this context, the government is promoting a combination of top-down and bottom-up approaches to boost national productivity.
The top-down strategy is implemented through industrialization, downstreaming, and large-scale investment, which are expected to generate demand for a more productive workforce.
Meanwhile, the bottom-up strategy is directed at direct intervention at the company level, especially medium-sized companies that create a significant number of jobs.
The Ministry of Manpower has stated that it will establish productivity clinics in 15 vocational training centers in 2026.
Currently, the ministry manages 42 vocational training centers that serve as facilities to improve the quality of Indonesian human resources.
The government has also prepared certification programs for productivity experts and productivity service operations targeting medium-sized enterprises.
To improve labor force productivity, the Ministry of Manpower has also implemented the National Internship Program as an instrument to strengthen industry-based skills.
This year, the program targets 100,000 participants in paid internships across state-owned enterprises (SOEs), private companies, and strategic institutions, as part of the bottom-up strategy to boost

productivity through direct work experience.

Unlike physical infrastructure development, productivity growth is largely achieved through non-physical investment, such as training, competency certification, improvements in work processes, and the strengthening of organizational culture.

Policy analyses consistently show that economic growth supported by productivity improvement is more resilient to external pressures than growth that relies on sectoral expansion.

Going forward, Indonesia's challenge lies not only in attracting investment, but also in ensuring that investment generates optimal added value through sustained improvements in labor productivity—an essential foundation for the country's economic competitiveness toward its Golden Indonesia 2045 vision.

Source: https://en.antaranews.com/news/396601/indonesias-productivity-gap-threatens-long-term-economic-growth

1. In growth theory, how does the quality of **Human Capital** (skills and education) affect the **Production Function**? Given the statistics in the article, why might an increase in the number of workers (L) fail to lead to a proportional increase in Indonesia's total output (Y)?
2. Based on the **Catch-up Effect (Convergence)**, Indonesia should theoretically grow faster than OECD countries. According to the article's discussion on the "structural nature" of Indonesia's challenges, what specific factors are preventing Indonesia from narrowing this productivity gap more quickly?
3. Define the difference between **extensive growth** (adding more inputs like labor and land) and **intensive growth** (improving productivity). Why does the Minister argue that intensive growth is the only way for Indonesia to sustainably reach its "Golden Indonesia 2045" vision of becoming a high-income country?

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. You are told that Country A experienced growth of real GDP per person of 4 percent per year throughout the 1900s. In view of other countries' experience, you would have to characterize Country A's growth as

a.	exceptionally high.
b.	moderately high.
c.	moderately low.
d.	exceptionally low.

____ 2. The level of real GDP person

a.	differs widely across countries, but the growth rate of real GDP per person is similar across countries.
b.	is very similar across countries, but the growth rate of real GDP per person differs widely across countries.
c.	and the growth rate of real GDP per person are similar across countries.
d.	and the growth rate of real GDP per person vary widely across countries.

____ 3. Which of the following countries had the highest growth rate over the last 100 or so years?

a.	Brazil
b.	Germany
c.	Canada
d.	United States

____ 4. Which of the following countries had the highest level of real GDP per person in 2006?

a.	Germany
b.	United Kingdom
c.	United States
d.	Japan

____ 5. The average amount of goods and services produced from each hour of a worker's time is called

a.	GDP.
b.	per capita GDP.
c.	productivity.
d.	technological knowledge.

____ 6. Dilbert's Incorporated produced 5,000,000 units of accounting software in 2004. At the start of 2005 the pointy-haired boss reduced total annual hours of employment from 10,000 to 8,000 and production was 4,800,000. These numbers indicate that productivity

a.	fell by 4%.
b.	fell by 20%.
c.	rose by 12%.
d.	rose by 20%.

____ 7. Dilbert's Incorporated produced 6,000,000 units of software in 2005. At the start of 2006 the pointy-haired boss raised employment from 10,000 total annual hours to 14,000 annual hours and production was 7,000,000 units. Based on these numbers what happened to productivity?

a.	It fell by about 16.7%.
b.	It stayed the same.
c.	It rose by about 16.7%.
d.	It rose by about 40%.

____ 8. Which of the following is human capital?

a.	textbooks
b.	hand held power tools
c.	understanding how to repair cars
d.	All of the above are correct.

____ 9. Which of the following statements is true?

a.	Natural resources per worker influence productivity only when those natural resources are renewable.
b.	The prices of most natural resources are stable or falling relative to other prices.
c.	Technology requires greater use of natural resources.
d.	The terms *human capital* and *technological knowledge* are used interchangeably.

____ 10. Suppose over the last year that the price of iron ore increased from $1,200 a ton to $1,300 a ton. Over the same time a measure of the overall price level increased from 168 to 187. The price of iron ore increased by

a.	less than inflation, and this means it became relatively less scarce.
b.	less than inflation, and this means it became scarcer.
c.	more than inflation, and this means it became scarcer.
d.	more than inflation, but this doesn't necessarily mean that it become scarcer.

____ 11. Human capital is

a.	the same thing as technological knowledge.
b.	the same thing as labor.
c.	the tools and equipment operated by humans.
d.	knowledge and skills that workers have acquired.

____ 12. If there are constant returns to scale, the production function can be written as

a.	$xY = 2xAF(L, K, H, N)$.
b.	$Y/L = A\,F(xL, xK, xH, xN)$.

c.	$Y/L = A\,F(1, K/L, H/L, N/L)$.
d.	L = AF(Y, K, H, N).

Scenario 25-1. An economy's production form takes the form $Y = AF(L, K, H, N)$.

____ 13. **Refer to Scenario 25-1**. In the production function, which variable would increase in value as technology improved?

a.	*A*
b.	*K*
c.	*H*
d.	*N*

____ 14. **Refer to Scenario 25-1**. If the production function has the constant-returns-to-scale property, then if we know the values of *A*, *K*/*L*, *H*/*L,* and *N*/*L*, we also know the value of

a.	output.
b.	labor productivity.
c.	*A*.
d.	All of the above are correct.

____ 15. Consider three imaginary countries. In Old York, saving amounts to $3,000 and consumption amounts to $7,000; in New Frank, saving amounts to $2,000 and consumption amounts to $8,000; and in Ganzee, saving amounts to $4,500 and consumption amounts to $10,500. The saving rate is

a.	higher in Old York than in Ganzee, and it is higher in Ganzee than in New Frank.
b.	higher in New Frank than in Ganzee, and it is higher in Ganzee than in Old York.
c.	higher in Ganzee than in New Frank, and it is the same in New Frank and Old York.
d.	higher in Old York than in New Frank, and it is the same in Old York and Ganzee.

____ 16. Over the period 1960-1991,

a.	South Korea had a higher growth rate than the United States because it had a higher ratio of investment to GDP.
b.	the United States had a higher growth rate than South Korea because it had a higher ratio of investment to GDP.
c.	South Korea had a higher growth rate than the United States even though it had a similar ratio of investment to GDP.
d.	the United States had a higher growth rate than South Korea even though it had a similar ratio of investment to real GDP.

____ 17. All else equal, if there are diminishing returns, then which of the following is true if a country increases its capital by one unit?

a.	Output will rise by more than it did when the previous unit was added.
b.	Output will rise but by less than it did when the previous unit was added.
c.	Output will fall by more than it did when the previous unit was added.
d.	Output will fall but by less then it did when the previous unit was added.

____ 18. All else equal, if there are diminishing returns, then if a country raised its capital by 100 units last year and by 100 units this year,

a.	the increase in output was greater for this year than last year.
b.	the increase in output was greater last year than this year.
c.	the increase in output is the same in both years.
d.	None of the above is necessarily correct.

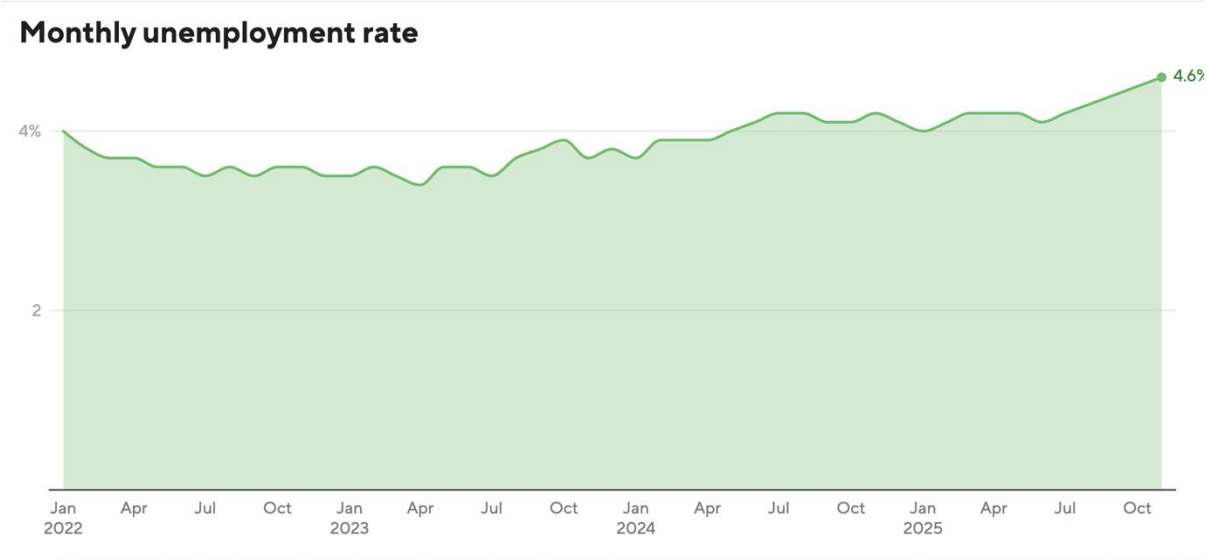
SHORT ANSWER

In addition to investment in physical and human capital, what other public policies might a country adopt to increase productivity?

TUT 3

CASE STUDY

Source: Bureau of Labor Statistics



Economists had forecast payroll gains of 40,000 jobs in November, according to a poll by FactSet.

The Labor Department on Tuesday also released partial employment data from October, which shows a loss of 105,000 jobs that month.

The drag on October employment was driven by a steep loss in the federal workforce, which shed 162,000 jobs that month. The decline is due to a wave of deferred resignations starting to show up in the payroll data, according to Samuel Tombs, chief U.S. economist at Pantheon Macroeconomics.

Job growth for August and September was also revised down by a collective 33,000.

The government's official employment data for October and November had been delayed due to the 43-day government shutdown, which ended in last month.

In the absence of federal data, economists at the Federal Reserve and on Wall Street had been monitoring alternative sources, which have signaled ongoing headwinds in the labor market. For instance, ADP earlier this month said private-sector employers in the U.S. cut 32,000 jobs in November, while outplacement firm Challenger, Gray & Christmas has tracked more than 1.1 million layoffs so far this year.

Economic uncertainty has led workers to stay put and employers to pull back on hiring, creating a difficult situation for job seekers, especially those early in their careers.

The labor market is slowly cooling, Federal Reserve Bank of New York president John C. Williams said in a speech Monday.

"I should emphasize that this has been an ongoing, gradual process, without signs of a sharp rise in layoffs or other indications of rapid deterioration," Williams said. But, he added, "Job growth has been anemic."

What it means

The unemployment rate ticked up to 4.6% in November — the highest since September 2021 — as more people entered the labor force to look for jobs, according to Adam Crisafulli, head of Vital Knowledge.

While the numbers offer a glimpse into the labor market during the government shutdown, experts say investors and government officials will likely view the data with some skepticism, given questions about the accuracy and completeness of data collection following the shutdown.

Indeed, the Federal Reserve may not place much emphasis on the November numbers at its January rate-setting meeting because of the data disruption, said Kay Haigh, global co-head of fixed income and liquidity solutions at Goldman Sachs Asset Management.

"The report on December's employment data, released in early January ahead of the next meeting, will likely be a much more meaningful indicator for the Fed when it comes to deciding the near-term policy trajectory," she said in an email.

Earlier this month, Federal Reserve Chair Jerome Powell said the central bank would likely view the October and November data with a "somewhat skeptical eye," given the data may be distorted.

The larger-than-expected rise in the unemployment rate could, however, raise concerns within the Fed, said Seema Shah, chief global strategist of Principal Asset Management, in an email.

The central bank lowered interest rates by 0.25 percentage points for the third consecutive time last week, citing the rise in the downside risks to employment.

Source: https://www.cbsnews.com/news/jobs-report-november-2025-economy-trump-hiring-bls/

1. How did the government shutdown and the resulting "data blackout" likely impact the sectoral shifts and the efficiency of job searches? Explain with the help of a graph
2. Distinguish between frictional and **structural unemployment** in the context of the "AI adjustment" mentioned in the article. If AI renders certain human skill sets obsolete while

creating a demand for others, does this represent a temporary "search" issue or a fundamental "mismatch"?

3. Explain how it is mathematically possible for the unemployment rate to rise even if the economy is technically adding jobs (64,000 in November). Does this rise necessarily indicate a "weak" economy, or could it represent a "re-entry" of discouraged workers?
4. How does the fear of "turnover" during uncertain times (as mentioned in the article) support the Efficiency Wage argument? Support your answer with a graph

Multiple Choice

Identify the choice that best completes the statement or answers the question.

____ 1. Wages in excess of their equilibrium level help explain

a.	structural unemployment but not the natural rate of unemployment.
b.	the natural rate of unemployment but not structural unemployment.
c.	both structural unemployment and the natural rate of unemployment.
d.	neither structural unemployment nor the natural rate of unemployment.

____ 2. Which of the following is *not* a reason that paying efficiency wages may increase a firm's profit?

a.	Efficiency wages increase worker health and therefore increase worker productivity.
b.	Efficiency wages decrease worker turnover and therefore decrease hiring and training costs.
c.	Efficiency wages decrease worker shirking and therefore increase worker productivity.
d.	Efficiency wages decrease a country's natural rate of unemployment and therefore increase its standard of living.

____ 3. Who of the following would be included in the Bureau of Labor Statistics' "unemployed" category?

a.	Cemal, a full-time student who is not looking for work
b.	Halim, who is on temporary layoff
c.	Zeynep, who has retired and is not looking for work
d.	All of the above are correct.

____ 4. Who is *not* included in the labor force by the Bureau of Labor Statistics?

a.	Kawanari, who is on temporary layoff
b.	Takuji, who has retired and is not looking for work
c.	Izumi, who does not have a job, but has applied for several in the last week
d.	None of the above is correct.

____ 5. A person who is counted as unemployed by the Bureau of Labor Statistics

a.	is also in the labor force.
b.	must have recently looked for work or be on temporary layoff.
c.	be at least 16 years old.
d.	All of the above are correct.

____ 6. The labor-force participation rate tells us the fraction of the population that

a.	is able to participate in the labor market.
b.	has ever been employed.
c.	has chosen to participate in the labor market.
d.	has chosen not to participate in the labor market.

____ 7. The labor-force participation rate is computed as

a.	(Employed Adult Population) 100.
b.	(Employed Labor Force) 100.
c.	(Labor Force Adult Population) 100.
d.	(Adult Population Labor Force) 100.

Table 28-2
2009 Labor Data for Wrexington

Number of adults	20,000
Number of adults who are paid employees	8,000
Number of adults who work in their own businesses	1,600
Number of adults who are unpaid workers in a family member's business	1,000
Number of adults who were temporarily absent from their jobs because of an earthquake	400
Number of adults who were waiting to be recalled to a job from which they had been laid off	200
Number of adults who do not have a job, are available for work, and have tried to find a job within the past four weeks	1,400
Number of adults who do not have a job, are available for work, but have not tried to find a job within the past four weeks	780
Number of adults who are full-time students	3,000
Number of adults who are homemakers or retirees	3,620

____ 8. **Refer to Table 28-2.** How many people were employed in Wrexington in 2009?

a.	9,600
b.	10,600
c.	11,000
d.	11,200

____ 9. Which of the following is *not* correct?

a.	Most people who become unemployed will soon find jobs.
b.	In an ideal labor market, wages would adjust to ensure that all workers are always fully employed.
c.	The unemployment rate occasionally falls to zero.
d.	There are always some workers without jobs, even when the overall economy is doing well.

____ 10. Meredith is looking for work as a computer programmer. Although her prospects are good, she hasn't yet taken a job. Julie is looking for work in a steel mill. Every time she shows up for an interview, there are more people looking for work than their are openings. Someone waiting in line with her tells her it has been that way for a long time.

a.	Meredith and Julie are both frictionally unemployed.
b.	Meredith and Julie are both structurally unemployed.
c.	Meredith is frictionally unemployed, and Julie is structurally unemployed.
d.	Meredith is structurally unemployed, and Julie is frictionally unemployed.

____ 11. From time to time, the demand for workers has risen in one region of the United States and fallen in another. This illustrates

a.	frictional unemployment created by efficiency wages.
b.	structural unemployment created by efficiency wages.
c.	frictional unemployment created by sectoral shifts.
d.	structural unemployment created by sectoral shifts.

____ 12. Which of the following is *not* a cause of frictional unemployment?

a.	the destruction of manufacturing jobs
b.	a worker leaving a job to find one with better benefits
c.	minimum-wage laws
d.	unemployment insurance

____ 13. Which of the following does *not* help reduce frictional unemployment?

a.	government-run employment agencies
b.	public training programs
c.	unemployment insurance
d.	All of the above help reduce frictional unemployment.

____ 14. Of the following groups, who is eligible for unemployment insurance benefits?

a.	the unemployed who quit their jobs
b.	the unemployed who were laid off because their previous employers no longer needed their skills
c.	the unemployed who were fired for cause
d.	the unemployed who just entered the labor force

____ 15. When a minimum-wage law forces the wage to remain above the level that balances supply and demand, there are

a.	more workers willing to work than there are jobs, so some workers are unemployed.
b.	fewer workers willing to work than there are jobs, so some workers are unemployed.
c.	more workers willing to work than there are jobs, so unemployment decreases.
d.	fewer workers willing to work than there are jobs, so unemployment decreases.

____ 16. If the minimum wage were currently above the equilibrium wage, than a decrease in the minimum wage would

a.	increase both the quantity demanded and the quantity supplied of labor.
b.	decrease both the quantity demanded and the quantity supplied of labor.
c.	increase the quantity of labor demanded but decrease the quantity of labor supplied.
d.	decrease the quantity of labor demanded but increase the quantity of labor supplied.

Short Answer

The table below uses data for the year 2003 provided by the Bureau of Labor Statistics and adjusted to be comparable to U.S. data. All values are in thousands. Fill in the blank entries in the table.

Country	Adult Population	Labor Force	Employed	Unemployed	Unemployment Rate	Labor-Force Participation Rate
Japan	109,474		62,510	3,500		
France		26,870		2,577		57.41
Germany	70,159	39,591			9.69	

TUT 4

CASE STUDY

Weak savings, declining investment constrain Kenya's job growth - report

KIPPRA Executive Director Eldah Onsomu speaking during the launch of the report.

The Kenya Institute for Public Policy Research and Analysis (Kippra) warns that weak domestic savings are widening structural constraints in the economy, limiting investment and contributing to slowing employment growth.

Kenya's ability to generate jobs is increasingly under pressure as the country struggles to save enough to support productive investments, a report has said.

Dubbed "Kenya Economic Report 2025" by the state-owned research firm, Kenya Institute for Public Policy Research and Analysis (Kippra), it warns that weak domestic savings are widening structural constraints in the economy, limiting investment and contributing to slowing employment growth.

The government mobilises savings through several avenues, including domestic revenue collection such as taxes on income, consumption, and corporate profits, as well as social contributions like pension schemes.

It can also raise funds by issuing government bonds and treasury bills to both institutional and retail investors, encouraging public participation in financing national projects.

Other avenues include leveraging state-owned enterprises' retained earnings, encouraging voluntary savings schemes, and creating development funds that pool resources from citizens, diaspora remittances, and private-sector contributions to support targeted investments and infrastructure projects.

Kippra reveals that the country's gross national savings have remained largely stable but subdued.

In 2024, savings declined marginally to 16.0 per cent of GDP from 16.1 per cent in 2023, remaining below the 2021 peak of 16.5 per cent.

This stagnation highlights the limited capacity of households, firms and the public sector to mobilise sufficient resources for investment-led growth.

Investment levels, however, have followed a clearer downward trend.

Total investment fell from 19.0 per cent of GDP in 2019 to 17.5 per cent in 2024, signalling reduced capital formation across key sectors of the economy.

"The declining trends in employment growth mirror the declining trends in investment," the report notes.

As a result, Kenya's savings-investment gap has narrowed in recent years, shrinking from 4.9 per cent of GDP in 2019 to 1.6 per cent in 2024.

However, Kippra cautions that this improvement does not reflect stronger savings performance.

Instead, it is largely the outcome of declining investment, raising concerns about the economy's future growth potential.

"The improvement in the savings-investment gap is due to a decline in investment rather than a substantial increase in savings."

1. If Kenyan households and firms continue to reduce their savings, explain what happens to the **Supply Curve** in the loanable funds market with a graph.

   - **Further question:**
   - The article mentions that Kenya's savings-investment gap narrowed from **4.9% in 2019 to 1.6% in 2024**. According to the KIPPRA report, why is this narrowing gap considered a **negative** sign for the economy rather than a positive one?

2. If the "savings-investment gap" forces interest rates to rise, explain how this change in the interest rate would likely affect the level of "capital formation" (investment) mentioned in the report.
   - Further question: Identify the specific percentage of GDP that Kenya's gross national savings fell to in **2024**, and compare it to the "peak" year mentioned in the article.

3. If Kenya successfully attracts a large amount of foreign capital, explain how this "capital inflow" would shift the **Supply of Loanable Funds** curve.
Further question: The report draws a direct parallel between the "declining trends in investment" and another specific economic metric. What is that metric, and what was the change in its growth rate between **2019 and 2024**?

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. If the government's expenditures exceeded its receipts, it would likely

a.	lend money to a bank or other financial intermediary.
b.	borrow money from a bank or other financial intermediary.
c.	buy bonds directly from the public.
d.	sell bonds directly to the public.

____ 2. Which of the following is correct?

a.	Some bonds have terms as short as a few months.
b.	Because they are so risky, junk bonds pay a low rate of interest.
c.	Corporations buy bonds to raise funds.
d.	All of the above are correct.

____ 3. Which of the following is *not* correct?

a.	If you buy a bond from a corporation, you can sell the bond to someone else before it matures.
b.	*Term* refers to the scheduling of periodic interest rate payments on a bond.
c.	A bond is an IOU.
d.	There are millions of different bonds in the U.S. economy.

____ 4. ABC Co. sells newly issued bonds. JLG Co. sells newly issued stocks. Which company is raising funds in financial markets?

a.	only ABC
b.	only JLG
c.	both ABC and JLG
d.	neither ABC nor JLG

____ 5. Suppose the government finds a major defect in one of a company's products and demands that the product be taken off the market. We would expect that the

a.	supply of existing shares of the stock and the price will both rise.
b.	supply of existing shares of the stock and the price will both fall.
c.	demand for existing shares of the stock and the price will both rise.
d.	demand for existing shares of the stock and the price will both fall.

____ 6. In the Coen Brothers' movie *The Hudsucker Proxy* the board of directors picks someone to run the company who they believe will make poor decisions. If things turn out as they plan,

a.	the price of a share of stock in the Hudsucker corporation should decline as the demand for shares falls.
b.	the price of a share of stock in the Hudsucker corporation should rise as the demand for shares rises.
c.	the price of a share of stock in the Hudsucker corporation should decline as the supply of existing shares falls.
d.	the price of a share of stock in the Hudsucker corporation should rise as the supply of existing shares rises.

____ 7. After a corporation issues stock, the stock

a.	can not be resold.
b.	can be resold only if the corporation wants to buy it back.
c.	can be resold on exchanges; the resale will raise additional funds for the corporation.
d.	None of the above are correct.

____ 8. Which of the following is both a financial institution and a financial intermediary?

a.	banks
b.	stock exchanges
c.	the bond market
d.	All of the above are correct.

____ 9. The primary advantage of mutual funds is that they

a.	always make a return that "beats the market."
b.	allow people with small amounts of money to diversify.
c.	provide customers with a medium of exchange.
d.	All of the above are correct.

____ 10. Suppose a closed economy had public saving of $3 trillion and private saving of $2 trillion. What are national saving and investment for this country?

a.	$5 trillion, $5 trillion
b.	$5 trillion, $2 trillion
c.	$1 trillion, $5 trillion
d.	$1 trillion, $2 trillion

____ 11. For a closed economy, GDP is $11 trillion, consumption is $7 trillion, taxes are $3 trillion and the government runs a surplus of $1 trillion. What are private saving and national saving?

a.	$4 trillion and $1 trillion, respectively
b.	$4 trillion and $5 trillion, respectively
c.	$1 trillion and $2 trillion, respectively
d.	$1 trillion and $1 trillion, respectively

____ 12. If in a closed economy Y = $11 trillion, which of the following combinations would be consistent with national saving of $2.5 trillion?

a.	C = $8 trillion, G = $.5 trillion
b.	C = $6.5 trillion, G = $3 trillion
c.	C = $8.5 trillion, G = $2 trillion
d.	C = $9 trillion, G = $.5 trillion

____ 13. The country of Yokovia does not trade with any other country. Its GDP is $30 billion. Its government purchases $5 billion worth of goods and services each year, collects $7 billion in taxes, and provides $3 billion in transfer payments to households. Private saving in Yokovia amounts to $5 billion. What are consumption and investment in Yokovia?

a.	$18 billion and $5 billion, respectively
b.	$21 billion and $4 billion, respectively
c.	$13 billion and $7 billion, respectively
d.	There is not enough information to answer the question.

____ 14. Other things the same, when the interest rate rises,

a.	people would want to lend more, making the supply of loanable funds increase.
b.	people would want to lend less, making the supply of loanable funds decrease.
c.	people would want to lend more, making the quantity of loanable funds supplied increase.
d.	people would want to lend less, making the quantity of loanable funds supplied decrease.

____ 15. Which of the following could explain an increase in the interest rate and the equilibrium quantity of loanable funds?

a.	The demand for loanable funds shifted rightward.
b.	The demand for loanable funds shifted leftward.
c.	The supply of loanable funds shifted rightward.
d.	The supply of loanable funds shifted leftward.

____ 16. Which of the following statements is correct?

a.	As a group, economists see no purpose in distinguishing between the nominal interest rate and the real interest rate.
b.	The interest rate that is usually reported is the nominal interest rate.
c.	If the nominal interest rate increases and the inflation rate remains unchanged, then the real interest rate decreases.
d.	All of the above are correct.

____ 17. Suppose that Congress were to institute an investment tax credit. What would happen in the market for loanable funds?

a.	The demand for loanable funds would shift left.
b.	The supply of loanable funds would shift left.
c.	The demand for loanable funds would shift right.
d.	The supply of loanable funds would shift right.

____ 18. Suppose government expenditures on goods and services and net taxes both decrease, and expenditures fall by more than net taxes. The effects of these changes on the budget deficit cause

a.	both the equilibrium interest rate and the equilibrium quantity of loanable funds to fall.
b.	both the equilibrium interest rate and the equilibrium quantity of loanable funds to rise.
c.	the equilibrium interest rate to rise and the equilibrium quantity of loanable funds to fall.
d.	the equilibrium interest rate to fall and the equilibrium quantity of loanable funds to rise.

____ 19. A change in the tax laws that increases the supply of loanable funds will have a bigger effect on investment when

a.	the demand for loanable funds is more elastic and the supply of loanable funds is more inelastic.
b.	the demand for loanable funds is more inelastic and the supply of loanable funds is more elastic.
c.	both the demand for and supply of loanable funds are more elastic.
d.	both the demand for and supply of loanable funds are more inelastic.

TUT 5

CASE STUDY

Central bank reduces intervention on OMO

The open market operations (OMO) have been more stable this month after the State Bank of Vietnam (SBV) reduced its intervention in the market.

Hanoi (VNS/VNA) - The open market operations (OMO) have been more stable this month after the State Bank of Vietnam (SBV) reduced its intervention in the market.

According to Saigon Securities Incorporation (SSI)'s research division, the OMO had seen strong fluctuations last month when the SBV consecutively injected a large amount of Vietnamese dong total 46 trillion VND through bill issues.

In August, the SBV changed its policy, starting to withdraw cash out of the market through the OMO. It withdrew more than 1 trillion VND through the bill channel in the first week of August.

However, SSI analysts noted, the SBV has still applied the interest rate bidding method in the OMO, with a floor rate of 3.5% per year, with an aim to keep the interbank lending interest rates at high levels.

Under the SBV's move, the interbank lending interest rate has hit a new high, accelerating from 0.3-0.5% per year to some 5% per year in the few weeks. On August 8, the interbank interest rate for dong loans was at 4.42% per year for overnight terms and 4.49% per year for one-week terms.

According to the analysts, the SBV's move to keep interest rates of dong loans in the interbank market at a high level is aimed at making a positive gap in lending interest rates between dong and US dollar loans, which will help reduce the need to hold the greenback among banks and indirectly cool down the foreign exchange rate.

By the end of last week, the USD/VND exchange rate on the interbank channel was traded at 23,389 VND per dollar, lower than the SBV's selling price of 23,400VND per dollar.

The listed exchange rate at Vietcom bank was at 23,250 VND and 23,530 VND per dollar for buying and selling, respectively, unchanged from last week.

Meanwhile, the exchange rate on the free market dropped sharply to 23,930 VND and24,020 VND per dollar for buying and selling, significantly narrowing the gap compared to the official market.

According to SSI, although the exchange rate has cooled down, the pressure is still relatively high as the market has not yet determined the time and extent of interest rate hikes by the US Federal Reserve (Fed).

SSI forecast the SBV might have to take stronger measures to stabilise the exchange rate at the end of the year when the Fed's dollar interest rate is expected to reach 3.5-3.75% per year in December.

Therefore, SSI suggested the interest rate of dong loans on the interbank channel needs to hit a higher level to reduce pressure on the exchange rate.

According to Tran Ngoc Bau, General Director of financial data provider WiGroup, the strong surge of the overnight interbank interest rate from 0.3-0.5% per year to some 5% per year in the past few weeks showed the SBV no longer maintained its loose monetary policy in the interbank market.

In addition, he said, the fact that the SBV actively withdrew money through the bill issue channel, and at the same time limited the credit growth quota for banks, can be seen as signs of tightening monetary policy.

The SBV's moves were mainly aimed to stabilise the USD/VND exchange rate, Bau said.

Vietcombank Securities Company (VCBS) also believes the SBV's recent interventions in the money market were aimed to reduce inflation and exchange rate pressures.

However, VCBS said, the fact that the Fed is still raising interest rates will make the greenback appreciate and maintain relative strength compared to other currencies. In this context, VCBS forecasts that the dong may depreciate by 3% against the dollar this year.

With the interbank interest rate, VCBS said the pressure from inflation has made the SBV act more cautiously and cannot maintain the abundant liquidity for banks.

However, VCBS expects the liquidity pressure in the interbank market will only happen in the short term and interest rates will gradually cool down at the end of the year, but are unlikely to return to low levels as per the average threshold of 2021./.

Source: https://en.vietnamplus.vn/central-bank-reduces-intervention-on-omo-post235647.vnp

1. Have you read the article yourself? What do the author mean by "injected" in the second paragraph? Which word can you replace it with? How do they do this (through which action?) Draw a diagram/graph to illustrate this situation.
2. In which paragraph do they mention an increase in the interest rate? How did this happen? Draw a graph of the money market and explain why the interest rate increase?
3. How much will the VND decrease its value against the dollar in this article? In which line did you find this information? How did they do it? How does this affect net export? And the Loanable funds market?

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. In an economy that relies upon barter,

a.	trade does not require a double coincidence of wants.
b.	scarce resources are allocated just as easily as they are in economies that do not rely upon barter.
c.	there is no item in the economy that is widely accepted in exchange for goods and services.
d.	All of the above are correct.

____ 2. Economists use the word "money" to refer to

a.	income generated by the production of goods and services.
b.	those assets regularly used to buy goods and services.
c.	the value of a person's assets.
d.	the value of stocks and bonds.

____ 3. Which type of money has intrinsic value?

a.	commodity money
b.	fiat money
c.	both commodity money and fiat money
d.	neither commodity money nor fiat money

____ 4. The Federal Reserve does all except which of the following?

a.	It controls the supply of money.
b.	It acts as a lender of last resort to banks.
c.	It makes loans to large business firms.
d.	It tries to ensure the health of the banking system.

____ 5. Which of the following does the Federal Reserve *not* do?

a.	conduct monetary policy
b.	act as a lender of last resort
c.	convert Federal Reserve Notes into gold
d.	serve as a bank regulator

____ 6. There is a

a.	short-run tradeoff between inflation and unemployment.
b.	short-run tradeoff between an increase in the money supply and inflation.
c.	long-run tradeoff between inflation and unemployment.
d.	long-run tradeoff between an increase in the money supply and inflation.

____ 7. Suppose that banks desire to hold no excess reserves, the reserve requirement is 5 percent, and a bank receives a new deposit of $1,000. This bank

a.	will increase its required reserves by $50.
b.	will initially see its total reserves increase by $1,000.
c.	will be able to make a new loan of $950.
d.	All of the above are correct.

____ 8. Suppose banks desire to hold no excess reserves. If the reserve requirement is 15 percent and if a bank receives a new deposit of $10, then this bank

a.	must increase its required reserves by $10.
b.	will initially see its total reserves increase by $15.
c.	will be able to make new loans up to a maximum of $8.50.
d.	All of the above are correct.

____ 9. If banks desire to hold no excess reserves, the reserve ratio is 10 percent, and a bank that was previously just meeting its reserve requirement receives a new deposit of $400, then initially the bank has a

a.	$400 increase in excess reserves and no increase in required reserves.
b.	$400 increase in required reserves and no increase in excess reserves.
c.	$360 increase in excess reserves and $40 increase in required reserves.
d.	$40 increase in excess reserves and $360 increase in required reserves.

____ 10. If the reserve ratio is 12.5 percent, the money multiplier is

a.	6.25.
b.	8.
c.	12.5.
d.	25.

Scenario 29-2.
The Monetary Policy of Tazi is controlled by the country's central bank known as the Bank of Tazi. The local unit of currency is the Taz. Aggregate banking statistics show that collectively the banks of Tazi hold 300 million Tazes of required reserves, 75 million Tazes of excess reserves, have issued 7,500 million Tazes of deposits, and hold 225 million Tazes of Tazian Treasury bonds. Tazians prefer to use only demand deposits and so all money is on deposit at the bank.

____ 11. Refer to **Scenario 29-2.** Suppose that the Bank of Tazi changes the reserve requirement ratio to 3 percent. Assuming that the banks still want to hold the same percentage of excess reserves what is the value of the money supply after the change in the reserve requirement ratio?

a.	9,375 million Tazes
b.	10,000 million Tazes
c.	12,500 million Tazes
d.	None of the above is correct to the nearest million salidos.

____ 12. Which of the following lists two things that both increase the money supply?

a.	lower the discount rate, raise the reserve requirement
b.	lower the discount rate, lower the reserve requirement
c.	raise the discount rate, raise the reserve requirement
d.	raise the discount rate, lower the reserve requirement

____ 13. Which of the following lists two things that both decrease the money supply?

a.	make open market purchases, raise the reserve requirement
b.	make open market purchases, lower the reserve requirement
c.	make open market sales, raise the reserve requirement
d.	make open market sales, lower the reserve requirement

____ 14. Suppose banks decide to hold more excess reserves relative to deposits. Other things the same, this action will cause the

a.	money supply to fall. To reduce the impact of this the Fed could lower the discount rate.
b.	money supply to fall. To reduce the impact of this the Fed could raise the discount rate.
c.	money supply to rise. To reduce the impact of this the Fed could lower the discount rate.
d.	money supply to rise. To reduce the impact of this the Fed could raise the discount rate.

____ 15. When the Fed conducts open-market sales,

a.	it sells Treasury securities, which increases the money supply.
b.	it sells Treasury securities, which decreases the money supply.
c.	it borrows from member banks, which increases the money supply.
d.	it lends money to member banks, which decreases the money supply.

____ 16. The Fed's primary tool to change the money supply is

a.	changing the discount rate.
b.	changing the reserve requirement.
c.	conducting open market operations.
d.	redeeming Federal Reserve notes.

Short Answer

1. What is meant by the term "lender of last resort?" In what circumstances might the Fed be a lender of last resort?

2. Explain why banks can influence the money supply if the required reserve ratio is less than 100 percent.

3. If the reserve ratio is 20 percent, how much money can be created from $100 of reserves? Show your work.

TUT 6

CASE STUDY

Understanding inflation in North Korea: Demand-pull and cost-push dynamics in a self-reliant economy

Looking forward, several plausible scenarios emerge, though most suggest a trajectory of sustained instability

By **Dr. Jungho Baek**

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September 15, 2025

North Korea's economy is experiencing a period of acute instability, characterized by rising consumer prices, currency depreciation, and increasing policy uncertainty. Inflation in particular has emerged as one of the most pressing challenges facing the regime, not only because it erodes real household incomes but also because it undermines the informal markets (*jangmadang*) that have sustained livelihoods since the collapse of the state distribution system in the 1990s. To understand what is happening, why it is happening, and what might lie ahead, it is helpful to analyze the phenomenon through the classic lenses of demand-pull and cost-push inflation. These frameworks clarify how domestic policy missteps and external shocks are interacting in ways that could destabilize both household welfare and the broader economic system.

What is happening with inflation?

Since late 2023, the North Korean government has implemented a comprehensive wage reform that has dramatically increased salaries for state-sector workers, in some cases by as much as fifty times. These sharp increases have injected large amounts of cash into the economy without any corresponding rise in productivity or output. From the perspective of demand-pull inflation, this represents a classic imbalance: more money is now circulating, but the supply of goods has not expanded in step. The immediate consequence is that households with access to wages have more nominal purchasing power, which pushes up demand for food and consumer items.

Simultaneously, the country is grappling with a currency crisis that has triggered cost-push inflation. The North Korean won has depreciated severely against the U.S. dollar, collapsing from roughly 8,000 won per dollar after the pandemic to over 40,000 won per dollar by mid-2025. This sharp devaluation has dramatically raised the cost of imported goods and commodities, which are critical inputs for both households and state industries. The price of rice has soared from approximately 4,000 won per kilogram to 20,000 won, while the cost of corn has tripled from 2,000 won to 6,000 won. Since grains represent around 70 percent of household consumption, such increases directly translate into severe hardship for the majority of citizens.

The interaction of these two forces—policy-induced demand surges and externally driven supply shocks—has created a volatile and unprecedented inflationary environment. Instead of strengthening the state sector, the reforms are eroding the purchasing power of wages, destabilizing the informal markets that provide most goods, and amplifying the vulnerabilities of an already fragile economy.

What are the underlying reasons?

The roots of this inflationary crisis lie in a misalignment between policy and economic fundamentals. At the core is the government's attempt to use wage hikes as a tool to revive the socialist economy and reassert control over labor. The official rationale was to lure workers away from the informal markets back into state factories and enterprises. However, without parallel increases in productivity—such as through the modernization of outdated machinery, the introduction of new technology, or the restoration of disrupted supply chains—higher wages translate into higher demand without corresponding increases in supply. In demand-pull terms, this is a textbook recipe for inflation: money incomes rise, but real goods do not, pushing up prices while eroding the real value of those very wages. For ordinary North Koreans, this effectively acts as an inflation tax, reducing their ability to purchase essentials and undermining the informal market system that sustains 70 percent of economic activity.

At the same time, structural supply constraints have intensified. North Korea's rigid adherence to self-reliance means that its economy lacks flexibility in production and is highly exposed to external shocks. The steep depreciation of the won against the U.S. dollar has inflated the price of imported goods, which are indispensable given the country's limited domestic production capacity. The depreciation—driven partly by the strengthening dollar and compounded by international sanctions and restricted trade with China—has led to soaring food and consumer prices. In such a context, supply cannot adjust quickly enough, embedding cost-push inflation into the economy.

Finally, the regime's broader misallocation of resources has worsened the inflationary spiral. Substantial state investments have been directed toward nuclear weapons, missile development, and massive construction projects, rather than into productive sectors such as agriculture or consumer manufacturing. This diversion of capital and labor reduces the economy's capacity to produce essential goods, amplifying shortages and making the state's wage policy even more inflationary. Thus, demand-side stimulus collides with supply-side rigidity, reinforcing the inflationary spiral and undermining the stability of markets.

What are the possible future scenarios?

Looking forward, several plausible scenarios emerge, though most suggest a trajectory of sustained instability. In the short term, the regime may continue to use Russian hard-currency inflows to finance wage payments and stabilize the exchange rate. This could temporarily mask the worst effects of inflation, but the benefits will be unevenly distributed: households with access to dollars or

state wages may cope, while the majority dependent on informal markets will face worsening poverty as prices for basic necessities remain at record highs.

In the medium term, the risks of stagflation are significant. If Russian support diminishes or sanctions tighten further, the government will lack the resources to maintain its inflated wage structure. In this scenario, nominal wages may remain high in official statistics, but their real value will plummet as the won continues to depreciate and cost-push pressures from import dependency persist. This would result in an economy characterized by both persistent inflation and stagnating output, exacerbating economic suffering.

There is also the possibility of historical repetition, echoing the 2009 currency reform. Then, state interventions wiped out household savings, triggered hyperinflation, and provoked rare public protests. The current experiment with wage hikes and forced re-centralization of the economy could lead to a similar outcome. If inflation accelerates beyond control, the regime may again resort to repression and scapegoating of traders, potentially dismantling the jangmadang system and intensifying hardship.

Finally, a scenario of policy confusion within the self-reliance framework appears likely. The government's ideological commitment to Juche economics clashes with its increasing dependence on external hard-currency inflows, particularly from Russia. As the won continues to depreciate against the dollar, and as the state attempts to sustain demand through wage increases while failing to bolster supply, contradictions in policy will deepen. Such reactive policymaking risks prolonging economic instability and eroding the resilience of both households and informal markets.

Conclusion

North Korea's inflation crisis is best understood as the outcome of simultaneous demand-pull and cost-push forces. On one side, wage hikes without productivity gains have spurred demand beyond the economy's capacity to supply, driving prices higher and imposing an inflation tax on households. On the other hand, currency depreciation, external sanctions, and chronic supply disruptions have led to cost-push inflation, resulting in persistently high prices in the economy. These pressures are intensified by the regime's misallocation of scarce resources toward military and prestige projects rather than productive sectors. The result is a deepening structural imbalance that undermines the informal markets responsible for most of the country's economic vitality. Looking ahead, North Korea may achieve only short-term stability by relying on external financial inflows; however, the more likely trajectory is one of stagflation, persistent inflation, or even a recurrence of past crises. Without a decisive pivot toward genuine supply-side reforms and greater international economic engagement, inflation will remain both a symptom and a driver of systemic fragility in the North Korean economy.

Source: https://www.dailynk.com/english/understanding-inflation-in-north-korea-demand-pull-and-cost-push-dynamics-in-a-self-reliant-economy/

Q1: Wage Reform and Rationale

- According to the article, by how much did the North Korean government increase state-sector wages starting in late 2023? What was the regime's primary "official rationale" for this drastic policy? Which graph do you use to illustrate this situation?

Q2: Currency and Commodity Prices

- Use the data in the article to describe the change in the value of the North Korean won against the USD from the post-pandemic period to mid-2025. How did this specifically impact the price of rice? What is the name of this incident? Where does it come from?

Q3: Demand-Pull vs. Cost-Push Dynamics

- The author describes a "classic imbalance" regarding demand-pull inflation in North Korea. Explain how the 2023 wage hike led to demand-pull inflation. Then, identify one factor mentioned in the text that is simultaneously causing cost-push inflation.

Q5: The "Inflation Tax"

- The author argues that for ordinary North Koreans, these reforms effectively act as an "inflation tax." Explain what this means in the context of the article. Who is most vulnerable to this "tax," and why?
- The article mentions North Korea's ideological commitment to *Juche* (self-reliance). How does the article suggest this ideology is currently being contradicted by the regime's financial reality? What "external factor" is temporarily masking the worst effects of inflation?

Multiple Choice

Identify the choice that best completes the statement or answers the question.

____ 1. In which of the following cases was the inflation rate 10 percent over the last year?

a.	One year ago the price index had a value of 110 and now it has a value of 120.
b.	One year ago the price index had a value of 120 and now it has a value of 132.
c.	One year ago the price index had a value of 126 and now it has a value of 140.
d.	One year ago the price index had a value of 145 and now it has a value of 163.

____ 2. When prices are falling, economists say that there is

a.	disinflation.
b.	deflation.
c.	a contraction.
d.	an inverted inflation.

____ 3. With the value of money on the vertical axis, the money supply curve is

a.	upward-sloping.
b.	downward-sloping.
c.	horizontal.
d.	vertical.

____ 4. The inflation tax refers to

a.	the revenue a government creates by printing money.
b.	higher inflation which requires more frequent price changes.
c.	the idea that, other things the same, an increase in the tax rate raises the inflation rate.
d.	taxes being indexed for inflation.

____ 5. When the money market is drawn with the value of money on the vertical axis, as the price level increases the quantity of money

a.	demanded increases.
b.	demanded decreases.
c.	supplied increases.
d.	supplied decreases.

____ 6. Suppose the money market, drawn with the value of money on the vertical axis, is in equilibrium. If the money supply increases, then at the old value of money there is an

a.	excess demand for money that will result in an increase in spending.
b.	excess demand for money that will result in a decrease in spending.
c.	excess supply of money that will result in an increase in spending.
d.	excess supply of money that will result in a decrease in spending.

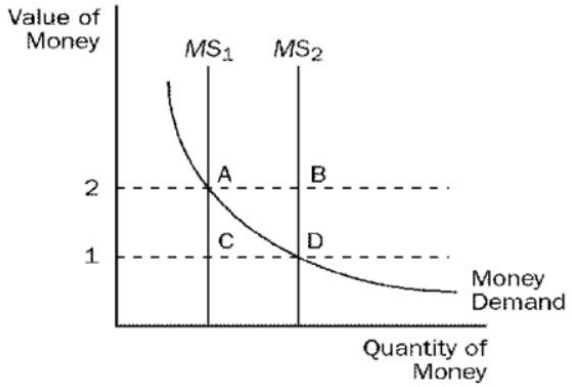
____ 7. If the Fed increases the money supply, then $1/P$

a.	falls, so the value of money falls.
b.	falls, so the value of money rises.
c.	rises, so the value of money falls.
d.	rises, so the value of money rises.

____ 8. When the money market is drawn with the value of money on the vertical axis, an increase in the money supply causes the equilibrium value of money

a.	and equilibrium quantity of money to increase.
b.	and equilibrium quantity of money to decrease.
c.	to increase, while the equilibrium quantity of money decreases.
d.	to decrease, while the equilibrium quantity of money increases.

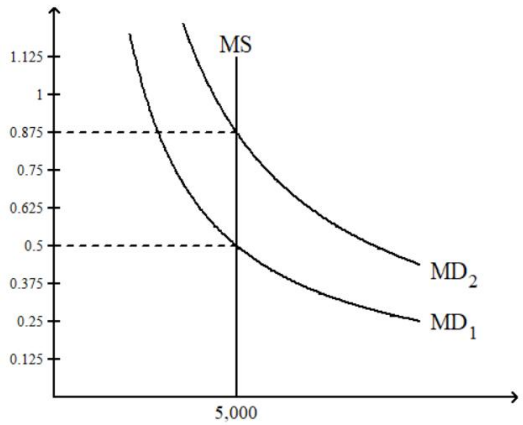
Figure 30-1



____ 9. **Refer to Figure 30-1**. If the money supply is MS_2 and the value of money is 2, then

a.	the value of money is lower than its equilibrium level.
b.	the price level is higher than its equilibrium level.
c.	the quantity of money demanded is greater than the quantity of money supplied.
d.	the quantity of money supplied is greater than the quantity of money demanded.

Figure 30-2. On the graph, MS represents the money supply and MD represents money demand. The usual quantities are measured along the axes.



____ 10. **Refer to Figure 30-2**. If the relevant money-demand curve is the one labeled MD_1, then

a.	when the money market is in equilibrium, one dollar purchases one-half of a basket of goods and services.
b.	when the money market is in equilibrium, one unit of goods and services sells for 2 dollars.
c.	there is an excess demand for money if the value of money in terms of goods and services is 0.375.
d.	All of the above are correct.

____ 11. Last year, Tealandia produced 50,000 bags of green tea, and tea was the only good Tealandia produced. Each bag sold at 4 units each of Tealandia's currency — the Leaf. Tealandia's money supply was 40,000. What was the velocity of money in Tealandia?

a.	20
b.	5
c.	1/20
d.	1/5

____ 12. If *Y* and *M* are constant, and *V* doubles, the quantity equation implies that the price level

a.	falls to half it's original level.
b.	doubles.
c.	more than doubles.
d.	does not change.

____ 13. Suppose over some period of time the money supply tripled, velocity fell by half, and real GDP doubled. According to the quantity equation the price level is now

a.	6 times its old value.
b.	3 times its old value.
c.	1.5 times its old value.
d.	0.75 times its old value.

____ 14. Suppose monetary neutrality holds and velocity is constant. A 5 percent increase in the money supply

a.	increases the price level by more than 5 percent.
b.	increases the price level by 5 percent.
c.	increases the price level by 5 percent.
d.	does not change the price level.

____ 15. The evidence from hyperinflations indicates that money growth and inflation

a.	are positively related, which is consistent with the quantity theory of money.
b.	are positively related, which is not consistent with the quantity theory of money.
c.	are not related in a discernible fashion, which is consistent with the quantity theory of money.
d.	are not related in a discernible fashion, which is not consistent with the quantity theory of money.

____ 16. Whitney puts money in a savings account at her bank earning 3.5 percent. One year later she takes her money out and notes that while her money was earning interest, prices rose 1.5 percent. Whitney earned a nominal interest rate of

a.	3.5 percent and a real interest rate of 5 percent.
b.	3.5 percent and a real interest rate of 2 percent.
c.	5 percent and a real interest rate of 3.5 percent
d.	5 percent and a real interest rate of 2 percent

____ 17. For a given real interest rate, a decrease in the inflation rate would

a.	decrease the after-tax real interest rate and so decrease saving.
b.	decrease the after-tax real interest rate and so increase saving.
c.	increase the after-tax real interest rate and so decrease saving.
d.	increase the after-tax real interest rate and so increase saving.

____ 18. Wealth is redistributed from creditors to debtors when inflation was expected to be

a.	high and it turns out to be high.
b.	low and it turns out to be low.
c.	low and it turns out to be high.
d.	high and it turns out to be low.

____ 19. If inflation is higher than what was expected,

a.	creditors receive a lower real interest rate than they had anticipated.
b.	creditors pay a lower real interest rate than they had anticipated.
c.	debtors receive a higher real interest rate than they had anticipated.
d.	debtors pay a higher real interest rate than they had anticipated.

Short Answer

1. Wages and prices are many times higher today than they were 30 years ago, yet people do not work a lot more hours or buy fewer goods. How can this be?

2. What is the inflation tax, and how might it explain the creation of inflation by a central bank?

3. Economists agree that increases in the money-supply growth rate increase inflation and that inflation is undesirable. So why have there been hyperinflations and how have they been ended?

TUT 7

CASE STUDY

Stronger solutions needed to boost domestic consumption demand

HÀ NỘI — Việt Nam needs to apply stronger solutions to boost domestic consumption demand to drive economic growth this year.

Domestic consumption is one of three drivers for growth, besides investment and export, and is the only pillar which finished the target last year with total retail sales of goods and services rising by 9.6 per cent, according to the General Statistics Office's report.

Public investment disbursement was estimated at 73.5 per cent of the plan while bottlenecks in the legal system, if not changed thoroughly, would continue to be a barrier in 2024.

Regarding trade, acceleration in the last months of 2023 could not help this pillar reach the finish line as import-export revenue fell by 6.6 per cent against 2022 to $683 billion, although the trade surplus was more than two times higher at $28 billion.

In the context of rising global uncertainties, aggregate demand was unlikely to recover to the level of the pre-pandemic period, so Việt Nam needs to take advantage of extensive and comprehensive cooperation to create momentum for 2024.

Lê Quốc Phương, former Deputy Director of the Việt Nam Industry and Trade Information Centre, said that the total retail sales of goods and services jumped quite high, by nearly 10 per cent, but the domestic aggregate demand was low. "The increase might be mainly from tourism and services, while consumption was not very positive," Phương said.

Low consumption demand stemmed from the economic recession and a decrease in average income, forcing consumers to tighten spending.

A survey by PwC last year found that Vietnamese consumers had become more cautious with their spending habits. Specifically, 63 per cent of surveyed consumers tended to cut unnecessary spending due to concerns over rising prices, 54 per cent cut spending on luxury products and 13 per cent cut spending on grocery and food products.

According to Vũ Vinh Phú, an expert in trade, it is necessary to introduce appropriate policies to stimulate domestic purchasing power.

The Government's policy of continuing reducing value added tax (VAT) by two percentage points to the end of June is a solution. However, the VAT reduction policy should be longer, together with bigger reductions for necessary products.

To stimulate domestic consumption, prices must be lower, Nguyễn Ngọc Hoà, President of HCM City Business Association said, adding that enterprises need support from the Government to implement demand stimulus activities.

Greater attention should be paid to developing the domestic market which plays a significant role in ensuring a firm foundation for the economy, economist Trần Đình Thiên said.

According to Lê Việt Nga, Deputy Director of the Domestic Market Department under the Ministry of Industry and Trade, the domestic market is always a foundation for Vietnamese businesses in all situations. "The domestic market needs to be further promoted with effective and practical demand stimulation programmes," she said. — VNS

Source: https://vietnamnews.vn/economy/1638941/stronger-solutions-needed-to-boost-domestic-consumption-demand.html

1. What are the main ideas of paragraph 1,2,3,4,5? Which problem does the economy face? What kind of gap do you think they mention in the article? Where do you think the government wants the output to be?

Draw the graph of the situation and illustrate the problem on the graph.

2. What was Mr. Phuong's point of view regarding consumption in comparison with other factors? Where did you find this information? Find a synonym for the word "to originate from" in the article. Why did the people tend to decrease their consumption? What kind of good do they consume less in your opinion?
3. What are some possible solutions does the government think of to solve the problem? How does that reflect in the formula of the total output? Can you draw a graph of the situation and explain how the solution may help close the gap we mentioned earlier in question 1?

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. During recessions investment

a.	falls by a larger percentage than GDP.
b.	falls by about the same percentage as GDP.
c.	falls by a smaller percentage than GDP.
d.	falls but the percentage change is sometimes much larger and sometimes much smaller.

____ 2. According to classical macroeconomic theory, changes in the money supply affect

a.	nominal variables and real variables.
b.	nominal variables, but not real variables.
c.	real variables, but not nominal variables.
d.	neither nominal nor real variables.

____ 3. The aggregate demand and aggregate supply graph has

a.	the price level on the horizontal axis. The price level can be measured by the GDP deflator.
b.	the price level on the horizontal axis. The price level can be measured by real GDP.
c.	the price level on the vertical axis. The price level can be measured by the GDP deflator.
d.	the price level on the vertical axis. The price level can be measured by GDP.

____ 4. The model of aggregate demand and aggregate supply explains the relationship between

a.	the price and quantity of a particular good.
b.	unemployment and output.
c.	wages and employment.
d.	real GDP and the price level.

____ 5. Other things the same, a fall in an economy's overall level of prices tends to

a.	raise both the quantity demanded and supplied of goods and services.
b.	raise the quantity demanded of goods and services, but lower the quantity supplied.
c.	lower the quantity demanded of goods and services, but raise the quantity supplied.
d.	lower both the quantity demanded and the quantity supplied of goods and services.

____ 6. In the context of aggregate demand and aggregate supply, the wealth effect refers to the idea that, when the price level decreases, the real wealth of households

a.	increases and as a result consumption spending increases. This effect contributes to the downward slope of the aggregate-demand curve.
b.	decreases and as a result consumption spending increases. This effect contributes to the upward slope of the aggregate-supply curve.
c.	increases and as a result households increase their money holdings; in turn, interest rates increase and investment spending decreases. This effect contributes to the downward slope of the aggregate-demand curve.
d.	decreases and as a result households increase their money holdings; in turn, interest rates increase and investment spending decreases. This effect contributes to the upward slope of the aggregate-supply curve.

____ 7. Other things the same, if the price level falls, people

a.	increase foreign bond purchases, so the supply of dollars in the market for foreign-currency exchange increases.
b.	increase foreign bond purchases, so the supply of dollars in the market for foreign-currency exchange decreases.
c.	decrease foreign bond purchases, so the supply of dollars in market for foreign-currency exchange increases.
d.	decrease foreign bond purchases, so the supply of dollars in the market for foreign-currency exchange decreases.

____ 8. As the price level rises,

a.	the exchange rate falls, so net exports fall.
b.	the exchange rate falls, so net exports rise.
c.	the exchange rate rises, so net exports fall.
d.	the exchange rate rises, so net exports rise.

____ 9. A decrease in U.S. interest rates leads to

a.	a depreciation of the dollar that leads to greater net exports.
b.	a depreciation of the dollar that leads to smaller net exports.
c.	an appreciation of the dollar that leads to greater net exports.
d.	an appreciation of the dollar that leads to smaller net exports.

____ 10. Other things the same, as the price level falls, the real value of a dollar

a.	rises, and interest rates rise.
b.	rises, and interest rates fall.
c.	falls, and interest rates rise.
d.	falls, and interest rates fall.

____ 11. If the dollar appreciates because of speculation or government policy

a.	aggregate demand shifts left. If other countries experience recessions, aggregate demand in the U.S. also shifts left.
b.	aggregate demand shifts left. If other countries experience recessions, aggregate demand in the U.S. shifts right.
c.	aggregate demand shifts right. If other countries experience recessions, aggregate demand in the U.S. shifts left.
d.	aggregate demand shifts right. If other countries experience recessions, aggregate demand in the U.S. also shifts right.

____ 13. Other things the same, if the long-run aggregate supply curve shifts right, prices

a.	and output both increase.
b.	and output both decrease.
c.	increase and output decreases.
d.	decrease and output increases.

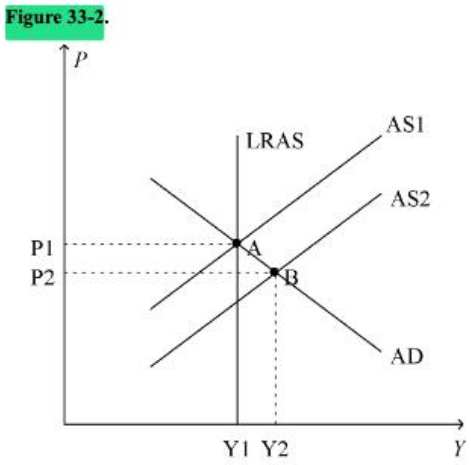
____ 14. The aggregate supply curve is upward sloping in

a.	the short and long run.
b.	neither the short nor long run.
c.	the long run, but not the short run.
d.	the short run, but not the long run.

____ 15. Other things the same, the aggregate quantity of output supplied will decrease if the price level

a.	is lower than expected so that firms believe the relative price of their output has increased.
b.	is lower than expected so that firms believe the relative price of their output has decreased.
c.	is higher than expected so that firms believe the relative price of their output has increased.
d.	is higher than expected so that firms believe the relative price of their output has decreased.

Figure 33-2.



____ 16. **Refer to Figure 33-2.** Point B represents

a.	a short-run equilibrium and a long-run equilibrium.
b.	a short-run equilibrium but not a long-run equilibrium.
c.	a long-run equilibrium but not a short-run equilibrium.
d.	neither a short-run equilibrium nor a long-run equilibrium.

The Stock Market Boom of 2014
Imagine that in 2014 the economy is in long-run equilibrium. Then stock prices rise more than expected and stay high for some time.

____ 17. **Refer to Stock Market Boom 2014.** In the short run what happens to the price level and real GDP?

a.	both the price level and real GDP rise.
b.	both the price level and real GDP fall.
c.	the price level rises and real GDP falls.
d.	the price level falls and real GDP rises.

Optimism
Imagine that the economy is in long-run equilibrium. Then, perhaps because of improved international relations and increased confidence in policy makers, people become more optimistic about the future and stay this way for some time.

____ 18. **Refer to Optimism.** What happens to the expected price level and what's the result for wage bargaining?

a.	The expected price level falls. Bargains are struck for higher wages.
b.	The expected price level falls. Bargains are struck for lower wages.
c.	The expected price level rises. Bargains are struck for higher wages.
d.	The expected price level rises. Bargains are struck for lower wages.

____ 19. Which of the following would increase the price level?

a.	an increase in the money supply.
b.	an increase in taxes.
c.	a decrease in the expected price level.
d.	a decrease in the natural rate of unemployment.

____ 20. When production costs rise,

a.	the short-run aggregate supply curve shifts to the right.
b.	the short-run aggregate supply curve shifts to the left.
c.	the aggregate demand curve shifts to the right.
d.	the aggregate demand curve shifts to the left.

Short Answer

1. Make a list of things that would shift the aggregate demand curve to the right.

2. Make a list of things that would shift the long-run aggregate supply curve to the right.

3. Suppose that a decrease in the demand for goods and services pushes the economy into recession. What happens to the price level? If the government does nothing, what ensures that the economy still eventually gets back to the natural rate of output?

TUT 8

CASE STUDY

Japan: Fiscal Consolidation and Gradual Monetary Policy Normalization Key to Sustaining Economic Resilience

SINGAPORE, March 4, 2025 – The Japanese economy is projected to grow at 1.3 percent in 2025, driven by sustained wage gains, strong business investments, and resilient performance in goods exports and tourism. While inflation has eased, it is expected to remain above the Bank of Japan's (BOJ) target, with CPI (excluding fresh food) inflation estimated to average 2.2 percent in 2025. Amid ongoing uncertainties surrounding the growth and inflation outlook, the BOJ should maintain a flexible, data-driven approach to monetary policy. Meanwhile, with public debt at elevated levels and rising fiscal pressures due to an aging population, stronger fiscal consolidation efforts will be needed to strengthen fiscal resilience.

These policy recommendations are highlighted in the 2024 Annual Consultation Report on Japan published today by the ASEAN+3 Macroeconomic Research Office (AMRO). The report is based on AMRO's Annual Consultation Visit to Japan which was completed on November 22, 2024, and data and information available up to December 13, 2024.

Recent developments and outlook

Japan's economic growth moderated to 0.1 percent in 2024 following the post-pandemic rebound. Although the economy contracted sharply in Q1 2024, growth rebounded over the subsequent three quarters, driven by strong consumption as wage hikes from the "Shunto" wage negotiations began to take effect. Wage increases are expected to remain high on account of a tight labor market and a shift in firm behavior. Corporate earnings are also expected to contribute to higher wages and increased capital investment in 2025, further boosting domestic demand. Goods exports and tourism are projected to remain robust, provided there is no sharp slowdown in major global economies.

While inflation slowed in 2024, CPI (excluding fresh food) inflation remained above the inflation target at 3.0 percent in December. Stripping out fresh food and energy, "core-core" CPI inflation was stable at 2.4 percent in the last two months of 2024 after increasing in successive months since July.

Amid a positive shift in firms' wage- and price-setting behavior, the BOJ terminated its yield curve control and negative-interest rate policy in March 2024, and renormalized its monetary policy framework using short-term interest rate as its main policy instrument.

Japan's current account surplus remained substantial at 4.8 percent of GDP in 2024, up from 3.8 percent in 2023. The larger surplus primarily reflects a narrower trade deficit and a higher primary income surplus. While automobile exports were negatively affected by factory closures related to safety certification issues, exports of semiconductors and semiconductor equipment surged reflecting the upswing in the tech cycle.

Credit growth expanded by over 3.3 percent in 2024, driven by resilient domestic demand and accommodative lending stance of banks. The overall banking system remains sound, supported by strong asset quality, adequate capital buffers, and robust profitability.

The fiscal deficit is projected to widen to 3.6 percent of GDP in FY2024 from 2.9 percent in FY2023 despite the gradual phase-out of some economic support packages. This increase reflects a sharper decline in revenue due to a temporary income tax cut, lower carryover revenue from the previous year, and higher spending from a supplementary budget amounting to JPY13.9 trillion. While public debt has been declining steadily after a significant increase during the COVID-19 pandemic, it remains high, at 240.6 percent of GDP in FY2024.

Risks, vulnerabilities, and challenges

Japan's macrofinancial outlook is skewed to the downside, reflecting substantial uncertainties, particularly from external factors. Key risks include a spike in global commodity prices triggered by escalating geopolitical tensions and a pronounced economic slowdown in other major economies.

On the domestic front, a slowdown in wage growth could undermine the BOJ's efforts to achieve its 2-percent inflation target. Conversely, a significant overshoot of inflation above its target could force a sharp tightening of monetary policy, straining households and businesses. Higher interest rates would also challenge fiscal sustainability by driving up interest payments on the government's substantial debt.

1. What is the author's attitude towards Japan's inflation in 2025? What happened to their economy from January to March in 2025? Why did it happen? How is that related to the graph of AD and AS? Find in the article a sign of a fiscal policy?
2. "On the domestic front, a slowdown in wage growth could undermine the BOJ's efforts to achieve its 2-percent inflation target." Explain this sentence with a graph
3. If Japan wants to achieve the same result with a monetary policy, how should they do it? Facilitate your answer with a graph

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. According to the liquidity preference theory, an increase in the overall price level of 10 percent

a.	increases the equilibrium interest rate, which in turn decreases the quantity of goods and services demanded.
b.	decreases the equilibrium interest rate, which in turn increases the quantity of goods and services demanded.
c.	increases the quantity of money supplied by 10 percent, leaving the interest rate and the quantity of goods and services demanded unchanged.
d.	decreases the quantity of money demanded by 10 percent, leaving the interest rate and the quantity of goods and services demanded unchanged.

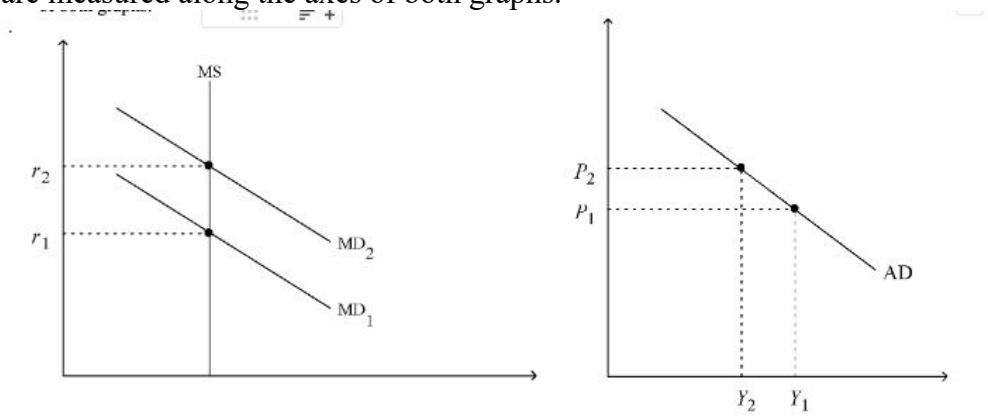
____ 2. In which of the following cases would the quantity of money demanded be *largest*?

a.	$r = 0.03$, $P = 1.4$
b.	$r = 0.03$, $P = 1.2$
c.	$r = 0.04$, $P = 1.2$
d.	$r = 0.06$, $P = 1.0$

____ 3. According to the theory of liquidity preference, which variable adjusts to balance the supply and demand for money?

a.	interest rate
b.	money supply
c.	quantity of output
d.	price level

Figure 34-2. On the left-hand graph, MS represents the supply of money and MD represents the demand for money; on the right-hand graph, AD represents aggregate demand. The usual quantities are measured along the axes of both graphs.



____ 4. **Refer to Figure 34-2**. A decrease in Y from Y_1 to Y_2 is explained as follows:

a.	The Federal Reserve increases the money supply, causing the money-demand curve to shift from MD_1 to MD_2; this shift of MD causes r to increase from r_1 to r_2; and this increase in r causes Y to decrease from Y_1 to Y_2.
b.	An increase in P from P_1 to P_2 causes the money-demand curve to shift from MD_1 to MD_2; this shift of MD causes r to increase from r_1 to r_2; and this increase in r causes Y to decrease from Y_1 to Y_2.
c.	A decrease in P from P_2 to P_1 causes the money-demand curve to shift from MD_1 to MD_2; this shift of MD causes r to increase from r_1 to r_2; and this increase in r causes Y to decrease from Y_1 to Y_2.
d.	An increase in the price level causes the money-demand curve to shift from MD_2 to MD_1; this shift of MD causes r to decrease from r_2 to r_1; and this decrease in r causes Y to decrease from Y_1 to Y_2.

____ 5. **Refer to Figure 34-2**. As we move from one point to another along the money-demand curve MD_1,

a.	the price level is held fixed at P_1.
b.	the interest rate is held fixed at r_1.
c.	the money supply is changing so as to keep the money market in equilibrium.
d.	the expected inflation rate is changing so as to keep the real interest rate constant.

____ 6. **Refer to Figure 34-2**. Assume the money market is always in equilibrium, and suppose $r_1 = 0.08$; $r_2 = 0.12$; $Y_1 = 13{,}000$; $Y_2 = 10{,}000$; $P_1 = 1.0$; and $P_2 = 1.2$. Which of the following statements is correct? When $P = P_2$,

a.	investment is lower than it is when $P = P_1$.
b.	nominal output is higher than it is when $P = P_1$.
c.	the expected rate of inflation is higher than it is when $P = P_1$.
d.	the velocity of money is higher than it is when $P = P_1$.

____ 7. According to liquidity preference theory, if the quantity of money demanded is greater than the quantity supplied, then the interest rate will

a.	increase and the quantity of money demanded will decrease.
b.	increase and the quantity of money demanded will increase.
c.	decrease and the quantity of money demanded will decrease.
d.	decrease and the quantity of money demanded will increase.

____ 8. Which of the following events would shift money demand to the right?

a.	an increase in the interest rate or an increase in the price level
b.	an increase in the interest rate, but not an increase in the price level
c.	an increase in the price level, but not an increase in the interest rate
d.	neither an increase in the interest rate nor an increase in the price level

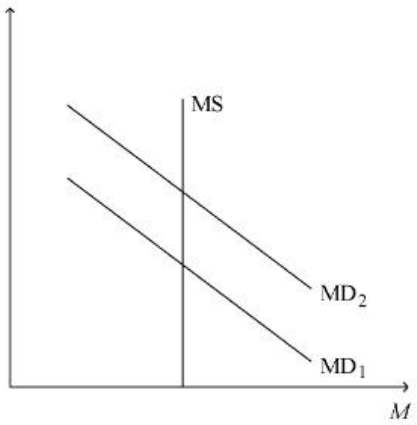
____ 9. Which of the following events would shift money demand to the left?

a.	an increase in the interest rate or an increase in the price level
b.	an increase in the interest rate, but not an increase in the price level
c.	an increase in the price level, but not an increase in the interest rate
d.	neither an increase in the interest rate nor an increase in the price level

____ 10. The multiplier for changes in government spending is calculated as

a.	*MPC*.
b.	1 - *MPC*.
c.	1/*MPC*.
d.	1/(1 - *MPC*).

Figure 34-4. On the figure, MS represents money supply and MD represents money demand.



____ 11. **Refer to Figure 34-4.** A shift of the money-demand curve from MD_2 to MD_1 is consistent with which of the following sets of events?

a.	The government cuts taxes, resulting in an increase in people's incomes.
b.	The government reduces government spending, resulting in a decrease in people's incomes.
c.	The Federal Reserve increases the supply of money, which decreases the interest rate.
d.	All of the above are correct.

____ 12. An increase in government spending initially and primarily shifts

a.	aggregate demand to the right.
b.	aggregate demand to the left.
c.	aggregate supply to the right.
d.	neither aggregate demand nor aggregate supply in either direction.

____ 14. If taxes

a.	increase, then consumption increases, and aggregate demand shifts rightward.
b.	increase, then consumption decreases, and aggregate demand shifts leftward.
c.	decrease, then consumption increases, and aggregate demand shifts leftward.
d.	decrease, then consumption decreases, and aggregate demand shifts rightward.

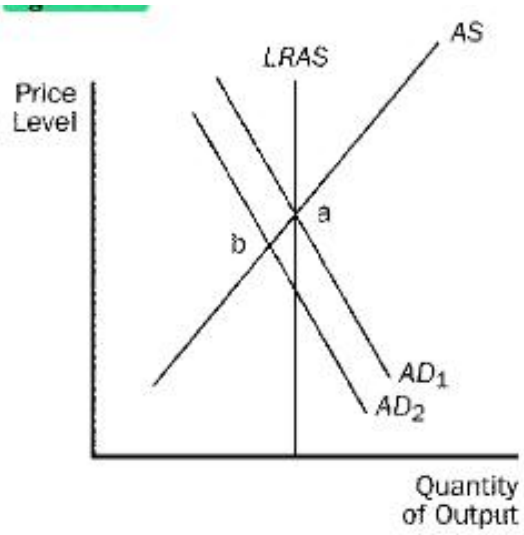
____ 17. If businesses and consumers become pessimistic, the Federal Reserve can attempt to reduce the impact on the price level and real GDP by

a.	increasing the money supply, which raises interest rates.
b.	increasing the money supply, which lowers interest rates.
c.	decreasing the money supply, which raises interest rates.
d.	decreasing the money supply, which lowers interest rates.

____ 18. Which of the following policies would be advocated by proponents of stabilization policy when the economy is experiencing severe unemployment?

a.	a decrease in the money supply
b.	a reduction in tax rates
c.	a decrease in government purchases
d.	None of the above is correct.

Figure 34-6.



____ 19. **Refer to Figure 34-6**. Which of the following is correct?

a.	Unemployment rises as the economy moves from point a to point b.
b.	Either fiscal or monetary policy could be used to move the economy from point b to point a.
c.	If the economy is left alone, then as the economy moves from point b to long-run equilibrium, the price level will fall farther.
d.	All of the above are correct.

Short Answer

1. Describe the process in the money market by which the interest rate reaches its equilibrium value if it starts above equilibrium.

2. Explain the logic according to liquidity preference theory by which an increase in the money supply changes the aggregate demand curve.

TUT 9

CASE STUDY

Export strength does not equal tech strength: Rethinking EU comparative advantage

Amid geopolitical shocks, the EU is revisiting its industrial strategy. However, traditional metrics often paint a misleading picture of Europe's underlying strength. This column uses a new approach to measure the EU's relative state of technology in different sectors. It shows that although EU manufacturing broadly outperforms the rest of the world in productivity terms, the industries in which it specialises are not always those in which it is relatively more productive. It argues that industrial policy must be rooted in productivity diagnostics and support should target sectors with strong relative state of technology but weak trade performance.

In the face of geopolitical shocks and intensifying global competition, the EU is revisiting its industrial strategy. Policymakers are particularly concerned about the erosion of Europe's technological edge vis-à-vis the US and China (Draghi 2024). Yet, how should one measure technological prowess in the first place?

Our recent research (di Mauro et al. 2025) suggests that traditional metrics like revealed comparative advantage – based on observed export shares – paint a misleading picture of Europe's underlying strength. Instead, we apply a methodology grounded in economic theory that purges trade data of distortions caused by input costs, market size, and other external factors. The resulting 'Ricardian' measure offers a more accurate view of the EU's relative state of technology. What we find is a sobering misalignment: Europe often specialises in industries where it does not hold a true productivity advantage, and vice versa.

Why revealed comparative advantage can be misleading

The Balassa index of revealed comparative advantage (BRCA) has long been used to infer a country's sectoral strengths. But BRCA is a noisy indicator. A country might export motor vehicles not because it is more productive at making them, but because it enjoys cheaper capital, looser regulations, or deeper supplier networks. That is, external competitiveness doesn't always reflect internal productivity.

To overcome this, we adopt a new model developed by Huang and Ottaviano (2024) that isolates technology-driven comparative advantage by controlling for:

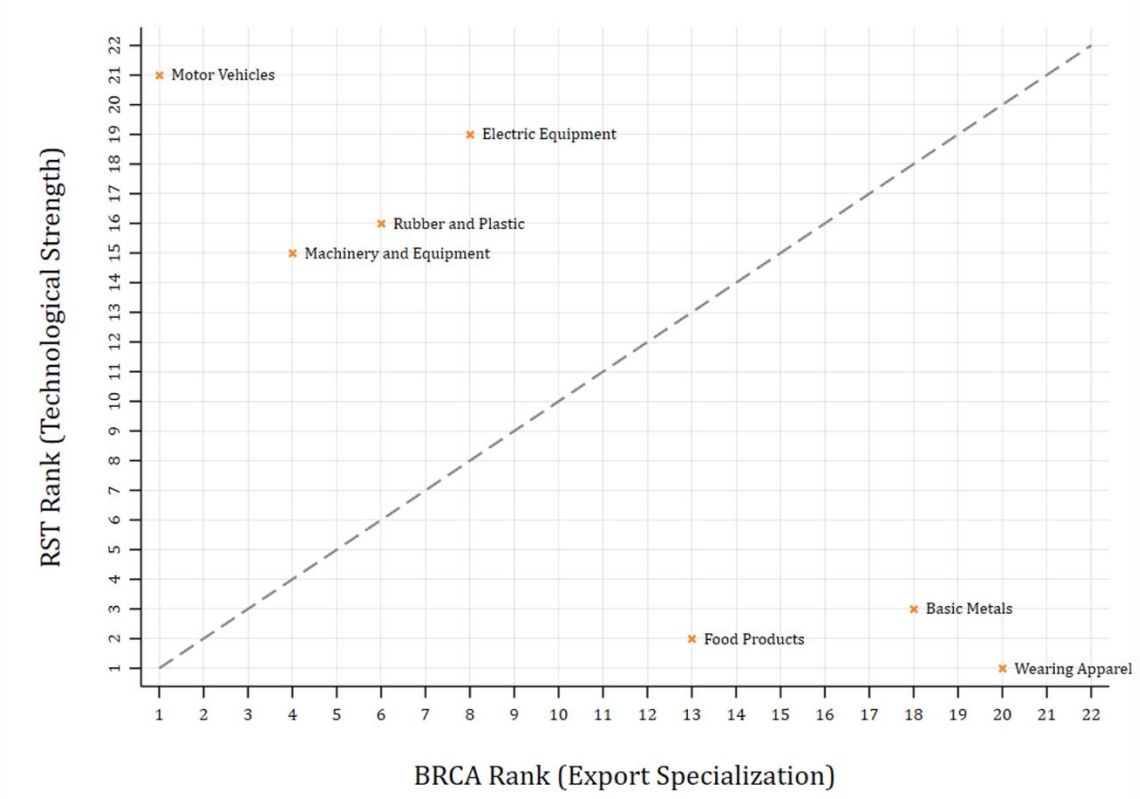
- Relative input prices (à la Heckscher-Ohlin)
- Market size and access (à la Krugman)
- Firm selection dynamics (à la Melitz)
- Trade frictions and heterogeneity

This model delivers a structural measure of a sector's relative state of technology, capturing its productivity net of market distortions.

Applying the model to the EU

We apply this framework to 16 EU countries using CompNet's micro-aggregated firm-level data and OECD input-output tables from 2010-2018. The model leverages data on export propensity, export intensity, market size, and firm heterogeneity to back out sectoral productivity.

Figure 1 Misalignment between export specialisation and technological strength



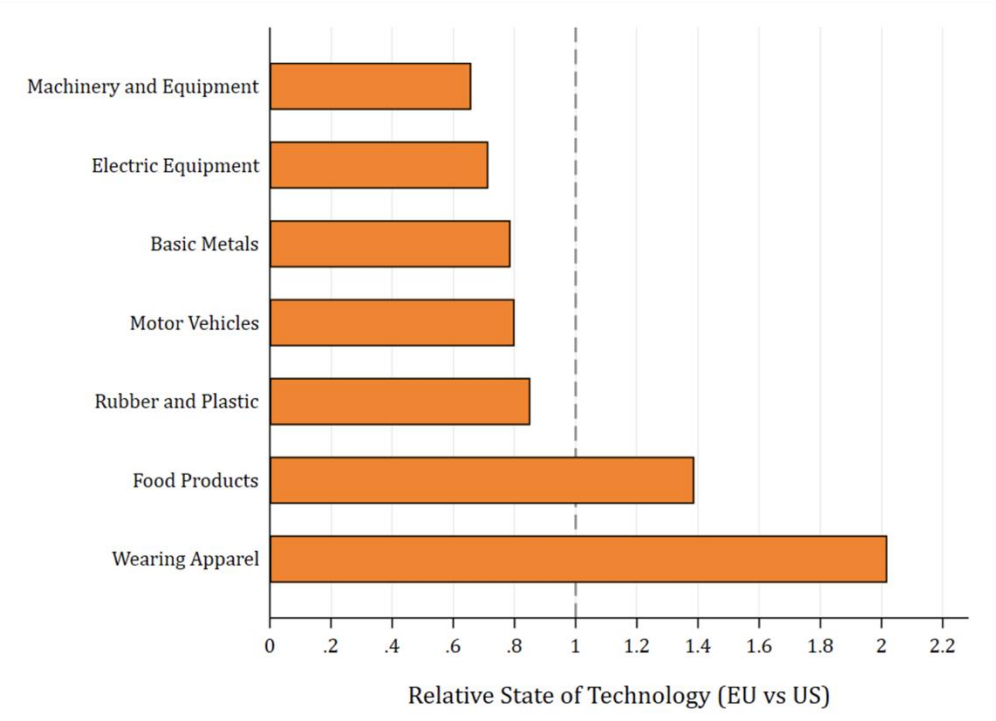
***Note*: Ranks for 2010-2018 averages (higher ranks are associated with smaller values). The 45-degree line shows perfect alignment between sectors' export specialisation (BRCA) and technological strength (RST). Sectors above the line are overrepresented in exports relative to their technological advantage.**

The key finding is that although EU manufacturing broadly outperforms the rest of the world in productivity terms, the industries it specialises in are not always those where it is relatively more productive. For example, wearing apparel shows a high technology advantage but weak export performance, whereas motor vehicles, by contrast, show strong export performance but lagging productivity.

The EU versus the US: A tougher benchmark

When we compare Europe to the US, the misalignment becomes starker. Europe's technological advantage erodes in key high-tech sectors like electrical equipment, machinery and equipment n.e.c., and motor vehicles. These are precisely the sectors Europe aims to bolster through in-dustrial policy.

Figure 2 EU vs US: Relative state of technology by sector



***Note*: Values greater than 1 indicate EU technological leadership. Europe underperforms the US in several high-tech sectors despite strong export performance.**

What drives the misalignment?

A decomposition shows that export specialisation depends on a complex interplay of:

- Input prices – lower costs can inflate export shares.
- Firm heterogeneity – more homogeneous firms correlate with higher observed revealed comparative advantage.
- Trade freeness – more open sectors attract more exporting firms, boosting revealed comparative advantage.
- Market size – larger domestic markets enable firms to scale and export more.

In short, favourable trade performance may reflect structural advantages other than productivity (e.g. subsidies, geographic clustering, or network effects).

Policy implications

Our findings have several important implications for EU policy. First, don't equate exports with excellence: export success should not be mistaken for technological strength, and policymakers risk promoting sectors that look competitive but are structurally weak. Second, industrial policy must be

rooted in productivity diagnostics; rather than blindly backing export winners, support should target sectors with strong relative state of technology but weak trade performance – these 'hidden champions' have untapped potential. Third, complement tech investment with eco-system reforms, since achieving comparative advantage requires more than innovation subsi-dies. The EU should ensure access to affordable inputs, deepen the Single Market to expand firm scale, streamline firm entry and exit to encourage competition, and invest in workforce skills and capital deployment.

Conclusion

As Europe ramps up its industrial strategy, it must distinguish between observed trade success and underlying productivity. Comparative advantage is about more than what you export – it's about why you export it. Our analysis shows that Europe's trade specialisation often deviates from its true technological strengths. For a smarter industrial policy, start with smarter metrics.

Source: https://cepr.org/voxeu/columns/export-strength-does-not-equal-tech-strength-rethinking-eu-comparative-advantage

1. What is the main idea of paragraph 1, 2, 3? What is a synonym for "originate from", "to be based on" or "to be developed from"? What is surprising about their findings? How do you identify if a country has the absolute advantage in producing something?
2. Which word in the passage means "inconsistency"? Which word means "tendency"? Which word means "disorder" or "messed up"? Which words means "to do better than" or "have more efficient outcome"?

   Draw an opportunity cost table between EU and US of electric equipment and food products?

Write a paragraph (4-5 sentences) describing figure 2?

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. When can two countries gain from trading two goods?

a.	when the first country can only produce the first good and the second country can only produce the second good
b.	when the first country can produce both goods, but can only produce the second good at great cost, and the second country can produce both goods, but can only produce the first good at great cost
c.	when the first country is better at producing both goods and the second country is worse at producing both goods
d.	Two countries could gain from trading two goods under all of the above conditions.

____ 2. Regan grows flowers and makes ceramic vases. Jayson also grows flowers and makes ceramic vases, but Regan is better at producing both goods. In this case, trade could

a.	benefit both Jayson and Regan.
b.	benefit Jayson, but not Regan.
c.	benefit Regan, but not Jayson.
d.	benefit neither Jayson nor Regan.

____ 3. Kelly and David are both capable of repairing cars and cooking meals. Which of the following scenarios is *not* possible?

a.	Kelly has a comparative advantage in repairing cars and David has a comparative advantage in cooking meals.
b.	Kelly has an absolute advantage in repairing cars and David has an absolute advantage in cooking meals.
c.	Kelly has a comparative advantage in repairing cars and in cooking meals.
d.	David has an absolute advantage in repairing cars and in cooking meals.

____ 4. Suppose that a worker in Freedonia can produce either 6 units of corn or 2 units of wheat per year, and a worker in Sylvania can produce either 2 units of corn or 6 units of wheat per year. Each nation has 10 workers. Without trade, Freedonia produces and consumes 30 units of corn and 10 units of wheat per year. Sylvania produces and consumes 10 units of corn and 30 units of wheat. Suppose that trade is then initiated between the two countries, and Freedonia sends 30 units of corn to Sylvania in exchange for 30 units of wheat. Freedonia will now be able to consume a maximum of

a.	30 units of corn and 30 units of wheat.
b.	40 units of corn and 30 units of wheat.
c.	40 units of corn and 20 units of wheat.
d.	10 units of corn and 40 units of wheat.

____ 5. Suppose that a worker in Freedonia can produce either 6 units of corn or 2 units of wheat per year, and a worker in Sylvania can produce either 2 units of corn or 6 units of wheat per year. Each nation has 10 workers. Without trade, Freedonia produces and consumes 30 units of corn and 10 units of wheat per year. Sylvania produces and consumes 10 units of corn and 30 units of wheat. Suppose that trade is then initiated between the two countries, and Freedonia sends 30 units of corn to Sylvania in exchange for 30 units of wheat. Sylvania will now be able to consume a maximum of

a.	30 units of corn and 30 units of wheat.
b.	40 units of corn and 30 units of wheat.
c.	40 units of corn and 20 units of wheat.
d.	10 units of corn and 40 units of wheat.

____ 6. Suppose that a worker in Agland can produce either 10 units of organic grain or 2 units of incense per year, and a worker in Zenland can produce either 5 units of organic grain or 15 units of incense per year. There are 20 workers in Agland and 10 workers in Zenland. Currently the two countries do not trade. Agland produces and consumes 100 units of grain and 20 units of incense per year. Zenland produces and consumes 50 units of grain and no incense per year. If each country made the decision to specialize in producing the good in which it has a comparative advantage, then the combined yearly output of the two countries would increase by

a.	30 units of grain and 100 units of incense.
b.	30 units of grain and 150 units of incense.
c.	50 units of grain and 90 units of incense.
d.	50 units of grain and 130 units of incense.

Table 3-1

Assume that Sardi and Tinaka can switch between producing corn and producing pork at a constant rate.

	Minutes Needed to Make 1	
	Bushel of Corn	Pound of Pork
Sardi	20	12
Tinaka	15	10

____ 7. **Refer to Table 3-1.** Assume that Sardi and Tinaka each has 360 minutes available. If each person divides his time equally between the production of corn and pork, then total production is

a.	10.5 bushels of corn and 16.5 pounds of pork.
b.	21 bushels of corn and 33 pounds of pork.
c.	35 bushels of corn and 22 pounds of pork.
d.	42 bushels of corn and 66 pounds of pork.

____ 8. **Refer to Table 3-1.** Which of the following combinations of corn and pork could Sardi produce in one 8-hour day?

a.	6 bushels of corn and 35 pounds of pork
b.	9 bushels of corn and 25 pounds of pork
c.	15 bushels of corn and 20 pounds of pork
d.	24 bushels of corn and 40 pounds of pork

____ 9. **Refer to Table 3-1.** What is Sardi's opportunity cost of producing one pound of pork?

a.	3/5 bushel of corn
b.	6/5 bushels of corn
c.	4/3 bushels of corn
d.	5/3 bushels of corn

Table 3-3

Assume that Zimbabwe and Portugal can switch between producing toothbrushes and producing hairbrushes at a constant rate.

	Machine Minutes Needed to Make 1	
	Toothbrush	Hairbrush
Zimbabwe	3	10
Portugal	5	6

____ 10. **Refer to Table 3-3.** Zimbabwe has an absolute advantage in the production of

a.	toothbrushes and a comparative advantage in the production of toothbrushes.
b.	toothbrushes and a comparative advantage in the production of hairbrushes.
c.	hairbrushes and a comparative advantage in the production of toothbrushes.
d.	hairbrushes and a comparative advantage in the production of hairbrushes.

Table 3-6

Assume that Hilda and Carlos can switch between producing quilts and producing dresses at a constant rate.

	Hours Needed To Make 1		Amount Produced in 90 Hours	
	Quilt	Dress	Quilts	Dresses
Hilda	30	10	3	9
Carlos	90	45	1	2

____ 11. **Refer to Table 3-6.** We could use the information in the table to draw a production possibilities frontier for Hilda and a second production possibilities frontier for Carlos. If we were to do this, measuring quilts along the horizontal axis, then

a.	the slope of Hilda's production possibilities frontier would be -3 and the slope of Carlos' production possibilities frontier would be -2.
b.	the slope of Hilda's production possibilities frontier would be -0.33 and the slope of Carlos' production possibilities frontier would be -0.5.
c.	the slope of Hilda's production possibilities frontier would be 0.33 and the slope of Carlos' production possibilities frontier would be 0.5.
d.	the slope of Hilda's production possibilities frontier would be 3 and the slope of Carlos' production possibilities frontier would be 2.

____ 12. **Refer to Table 3-6.** We could use the information in the table to draw a production possibilities frontier for Hilda and a second production possibilities frontier for Carlos. If we were to do this, measuring dresses along the horizontal axis, then

a.	the slope of Hilda's production possibilities frontier would be -3 and the slope of Carlos' production possibilities frontier would be -2.
b.	the slope of Hilda's production possibilities frontier would be -0.33 and the slope of Carlos' production possibilities frontier would be -0.5.
c.	the slope of Hilda's production possibilities frontier would be 0.33 and the slope of Carlos' production possibilities frontier would be 0.5.
d.	the slope of Hilda's production possibilities frontier would be 3 and the slope of Carlos' production possibilities frontier would be 2.

Table 3-9

Barb and Jim run a business that sets up and tests computers. Assume that Barb and Jim can switch between setting up and testing computers at a constant rate. The following table applies.

	Minutes Needed to		Number of Computers Set Up or Tested in a 40-Hour Week	
	Set Up 1 Computer	Test 1 Computer	Computers Setup	Computers Tested
Barb	48	?	50	40
Jim	30	40	80	60

____ 13. **Refer to Table 3-9.** Barb has an absolute advantage in

a.	both setting up and testing computers and a comparative advantage in setting up computers.
b.	both setting up and testing computers and a comparative advantage in testing computers.
c.	neither setting up nor testing computers and a comparative advantage in setting up computers.
d.	neither setting up nor testing computers and a comparative advantage in testing computers.

Table 3-11

Assume that Falda and Varick can switch between producing wheat and producing cloth at a constant rate.

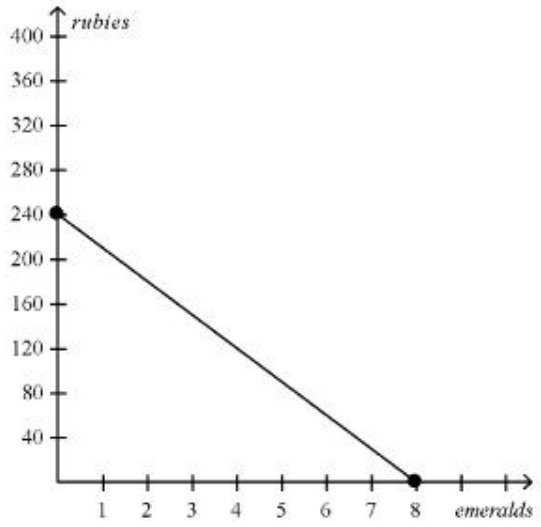
	Quantity Produced in 1 Hour	
	Bushels of Wheat	Yards of Cloth
Falda	8	12
Varick	6	15

____ 14. **Refer to Table 3-11.** Falda's opportunity cost of one bushel of wheat is

a.	2/3 yard of cloth and Varick's opportunity cost of one bushel of wheat is 2/5 yard of cloth.
b.	2/3 yard of cloth and Varick's opportunity cost of one bushel of wheat is 5/2 yards of cloth.
c.	3/2 yards of cloth and Varick's opportunity cost of one bushel of wheat is 2/5 yard of cloth.
d.	3/2 yards of cloth and Varick's opportunity cost of one bushel of wheat is 5/2 yards of cloth.

Figure 3-2

Peru's Production Possibilities Frontier

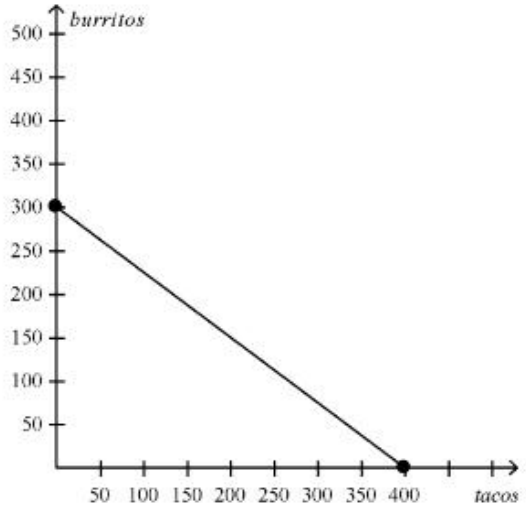


____ 15. **Refer to Figure 3-2**. If the production possibilities frontier shown is for 40 hours of production, then how long does it take Peru to make one ruby?

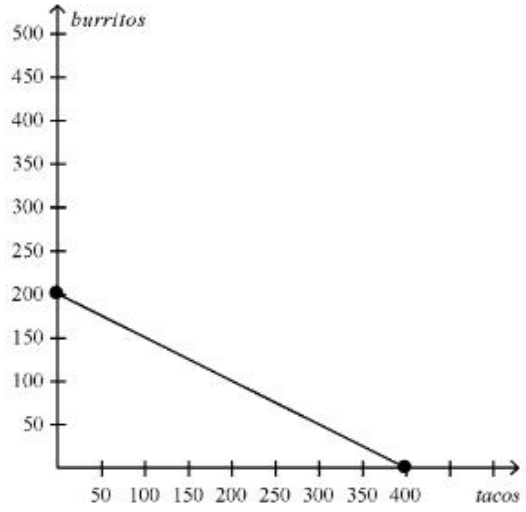
a.	1/6 hour
b.	1/5 hour
c.	5 hours
d.	6 hours

Figure 3-3

Bob's Production Possibilities Frontier



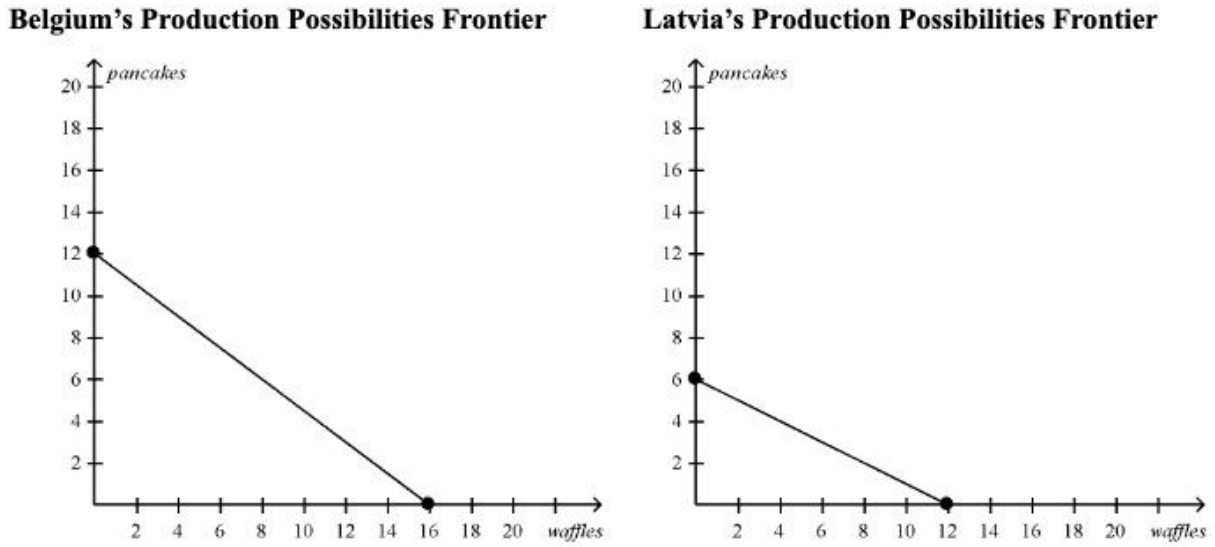
Enid's Production Possibilities Frontier



____ 16. **Refer to Figure 3-3.** Enid has an absolute advantage in the production of

a.	burritos and a comparative advantage in the production of tacos.
b.	burritos and a comparative advantage in the production of burritos.
c.	neither good and a comparative advantage in the production of tacos.
d.	neither good and a comparative advantage in the production of burritos.

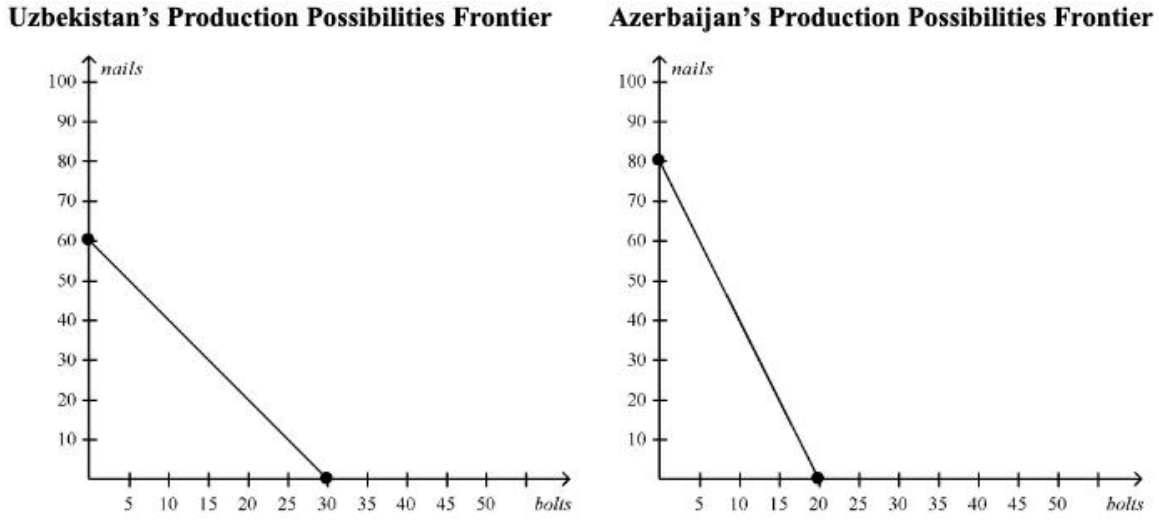
Figure 3-8



____ 17. **Refer to Figure 3-8.** Belgium should specialize in the production of

a.	waffles and import pancakes.
b.	pancakes and import waffles.
c.	both goods and import neither good.
d.	neither good and import both goods.

Figure 3-9



____ 18. **Refer to Figure 3-9**. If the production possibilities frontiers shown are each for two days of production, then which of the following combinations of bolts and nails could Uzbekistan and Azerbaijan together make in a given 2-day production period?

a.	12 bolts and 120 nails
b.	24 bolts and 96 nails
c.	38 bolts and 50 nails
d.	44 bolts and 24 nails

Short Answer

1. Explain the difference between absolute advantage and comparative advantage. Which is more important in determining trade patterns, absolute advantage or comparative advantage? Why?

2. Gary and Diane must prepare a presentation for their marketing class. As part of their presentation, they must do a series of calculations and prepare 50 PowerPoint slides. It would take Gary 10 hours to do the required calculation and 10 hours to prepare the slides. It would take Diane 12 hours to do the calculations and 20 hours to prepare the slides.

a.	How much time would it take the two to complete the project if they divide the calculations equally and the slides equally?
b.	How much time would it take the two to complete the project if they use comparative advantage and specialize in calculating or preparing slides?
c.	If Diane and Gary have the same opportunity cost of $5 per hour, is there a better solution than for each to specialize in calculating or preparing slides?